# Measuring the Task Value of Precision Upgrades in Closed-Loop Agents

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## Abstract

Evaluations of low-precision agents usually measure drift from BF16. A deployment that can fall back to BF16 needs the task value of a specific upgrade under the policy that continues. We restore a decision state, audit exact replay on sampled states, replace the current action, the continuation, or both with BF16, and count rescues and damages. On Qwen3 agents under simulated quantization in $\tau^2$-bench, replacing a W4 agent's first tool-structure-divergent action with BF16's changes 15.5–20.9% of outcomes in both directions; the pooled net is $+0.7$ points (95% interval over new tasks $[-2.1, +3.4]$). On held-out telecom tasks, the pre-registered net estimate is $-4.6$ points (90% interval $[-7.2, -2.0]$); the frozen decision rule had no category for a negative effect, so it classifies the result as inconclusive. Two reference interventions also change outcomes: a second same-bit-width checkpoint, whose rescues and damages are both resolved above zero, and reversing tool-argument key order, which leaves the tool operation identical (exploratory). At common states, BF16's extra action value over that checkpoint is not resolved, and equivalence is not shown. Outcome sensitivity is therefore not by itself evidence that an upgrade helps. A frozen uncertainty trigger selects outcome-changing decisions without a reliable net gain.

## 1 Introduction

A quantized LLM agent (Q) is cheaper to serve than its BF16 reference (B), and a deployment can keep B as a fallback (Li et al., 2026c). Evaluations usually report how far Q drifts from B. A fallback needs the task value of one specified upgrade: at a decision where B would act differently, B can **replace only this step, take over from the next step, or both**. A more sensitive model, a more different action or a more changeable outcome does not by itself show that an upgrade helps, and average success does not answer the question: on our checkpoints, Q's success minus B's ranges from $-0.2$ to $+4.2$ pp across four W4 configurations (a configuration is one model under one quantization setting), and equivalence within $\pm 3$ pp is not established in any of them (Appendix A.2).

Figure 1 shows one decision from our data. In an airline task, Q looks up a reservation id that does not exist; B looks up the user's profile. We froze the state and ran all four combinations of {B's or Q's action} $\times$ {B or Q continuing}. Only B's action followed by B's continuation succeeds. When B supplies only this step and Q continues, the task fails; when Q's failed lookup stands and B takes over, it fails too. Here B's action is worth a success only if B also continues, and no evaluation of the action alone can show that. The case illustrates the measurement distinction; it is not evidence for a mechanism.

When a forked trajectory is continued with another model (Gonuguntla, 2026), that model writes both the next action and everything after it, so the two effects cannot be separated. We cross the source of the current action with the model that executes the rest of the episode, at a frozen state whose exact replay we audit (§3.3). We read the results against two reference interventions: a second W4 checkpoint of the same model (Q′) in place of B, and an argument key-order change that leaves the tool operation identical. We study Qwen3-4B and Qwen3-8B (Qwen Team, 2025) under simulated weight and KV-cache quantization in $\tau^2$-bench (Yao et al., 2024; Barres et al., 2025); BFCL multi-turn (Patil et al., 2025) and ALFWorld (Shridhar et al., 2021) (other environments) and Mistral-Small-3.2-24B (Mistral AI, 2025) (a second model family, forked at sampled $\tau^2$-bench states) are scope checks (Table 2).

**Contribution.** Our contribution is an evaluation protocol for the task value of a precision upgrade, and two reference experiments that show why sensitivity-based inferences about that value fail. It

makes three targets explicit:

- *Measurement target.* Replacing the current action, taking over the continuation and replacing both are different interventions, each measured on the same state by its rescues and damages; a gain measured for one is not reused for another.
- *Interpretation reference.* A same-bit-width checkpoint produces two-sided consequences, and a representation change that keeps the tool operation identical also changes outcomes (rescues and damages both observed; exploratory). Flips are read against these references, not subtracted from B's.
- *Selection target.* Selecting decisions that change outcomes and selecting upgrades with positive net value are separate goals; we test a frozen rule against random and oracle selection with the same budget.

Valuing a step under the continuing policy and factorial forking are standard (Setlur et al., 2024); what we add is the three contrasts applied to precision upgrades at shared states, the two references, and the replay audit.

**Findings.**

- **An upgrade has two-sided consequences** (§4). Replacing a W4 agent's first tool-structure-divergent action with B's changes 15.5–20.9% of task outcomes, with rescues and damages of similar size; pooled over the four W4 configurations the net is $+0.7$ pp (95% over new tasks $[-2.1, +3.4]$). In post hoc analyses, testing the action alone ($G$) and testing the joint replacement alone ($F$) each miss rescues that the other counts. A pre-registered telecom holdout again shows rescues and damages. Its net estimate is $-4.6$ pp; the frozen decision rule, which had no negative category, classifies it as inconclusive, and both pre-registered secondary checks passed.
- **Flips alone do not diagnose a need for more bits** (§5). Under a protocol frozen before any outcome, a second checkpoint at the same bit width also produces two-sided flips and continuation dependence (turnover 17.7% against 16.8% for B on the same blocks; equivalence to B not shown). At common states B's extra action value over it is not established, nor is equivalence. Reversing tool-argument key order, which leaves the operation identical, changes 6.7% of outcomes on 300 eligible states, not distinguished from 9.0% for B's action (exploratory).

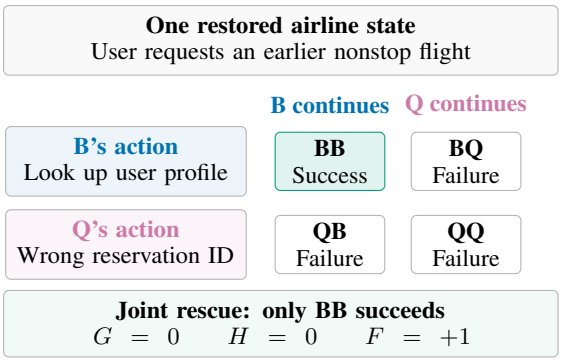



Figure 1: Four branches at one restored state (Qwen3-4B, AWQ-style W4; airline task 31, decision 1 of 12). Only the joint replacement succeeds; action-only and continuation-only evaluations miss this rescue. The case illustrates the measurement distinction, not a mechanism.

- **Selecting sensitive decisions is not selecting valuable upgrades, for one frozen rule** (§6). A frozen Q-only margin trigger selects more outcome-changing decisions than random at two of three escalation rates, rescues and damages alike. Its net gains ($+4$, $+2$ and $+5$ of 1020 states) are not reliably above random expectations ($+0.21$, $+0.98$, $+2.38$; no rate survives Holm correction); a hindsight oracle with the same budget nets $+34$, $+38$ and $+38$. The rule was tested only on later blocks of the development tasks.

## 2 Related Work

**Quantized agents compared episode by episode.** *Flat Score, Amplified Failures* (Jang et al., 2026) is closest in setting: on $\tau^2$-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, paired tasks flip in both directions, and recovery is equally frequent at both precisions. Its episodes are sampled independently, so the precision of the failing action and that of the recovering controller change together; we vary the two separately at a fixed state. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

**Forks and counterfactual replay.** *The Replay Gap* (Gonuguntla, 2026) forks live SWE-bench Verified trajectories for per-step model switching; its swap replaces the next action and the continuation together (our "both" contrast), and its control holds the model fixed, ours the bit width. Replacing one

action at a restored state and letting the original policy continue (our "this step only" contrast) locates decisive errors (Zhang et al., 2025), attributes failures (Shah, 2026) and yields training signal (Li et al., 2026a,b); the factorial design itself is standard.

**Precision routing, escalation and their signals.** DMR (Li et al., 2026c) learns a per-step BF16/W4 router from KL supervision and episode reward; sensitivity has been localised to phases (Lu et al., 2026), task types (Zhang et al., 2026) and high-entropy, high-KL tokens (Lotfi et al., 2026). Agents also defer single decisions when uncertain (Piatrashyn et al., 2026), learn cost-aware escalation (Lei and Huang, 2026), or are routed per step against execution-verified labels (Yang et al., 2026). Continuing another model's trajectory has its own cost (Ganz et al., 2026); a policy's confidence does not reliably locate consequential steps under executed replay (Zhang, 2026); and KL measures how much a quantized model disagrees, not in which direction (Nikolić et al., 2026). We propose no router and test no learned one; we add separate action, continuation and joint replacement on the same states, with rescue and damage counts and same-precision references.

**Evaluation noise and user simulators.** Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

# 3 Measuring the Task Value of an Upgrade

## 3.1 Setting

**Tasks and agents.** We use $\tau^2$-bench in text mode: a development panel of 58 retail and 27 airline tasks plus 80 telecom development tasks, and a holdout of 34 telecom tasks that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8); a second family, Mistral-Small-3.2-24B, is a scope check (Table 2). One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success is the final-database match with normal termination, without all of $\tau^2$-bench's official communication checks; telecom uses the official task-basis success. BFCL multi-turn and ALFWorld have no user simulator and run once per item (Appendix A).

**Quantization and serving.** Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. A *configuration* is one model under one quantization setting. The main configurations use GPTQ (Frantar et al., 2023) and AWQ-style scaling (Lin et al., 2024) at W4 (group size 128), calibrated on BF16 agent trajectories of 57 retail and airline tasks disjoint from the development tasks, and an INT8 KV cache (KV8). Stress configurations (GPTQ-W3, RTN-W3, rotated KV4) and QuaRot, SpinQuant and KIVI (Ashkboos et al., 2024; Liu et al., 2025, 2024) are scope checks (Table 2; Appendix Table 5). B and Q run on vLLM (Kwon et al., 2023) with batch-invariant kernels (He and Thinking Machines Lab, 2025) and deterministic compilation in separate server processes, so no prefix cache is shared. A *block* is one closed-loop run of Q over the task pool under one prompt nonce (a random session-identifier line in the system prompt); blocks are the replicates, and a *unit* is one task in one block.

## 3.2 Four branches on one state

**Frozen state.** Before Q's decision $t$ we restore the state $s_t$: the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q generate their actions from identical prompt token ids, and the state is written to disk before any branch runs.

**Four branches.** $Y_{\mathrm{XY}} \in \{0,1\}$ is task success when the action at $s_t$ comes from X and model Y executes the rest of the episode, X, Y $\in \{\mathrm{B},\mathrm{Q}\}$ (Figure 1). The action is executed by the real environment or answered by the real simulator; all four branches share the remaining budget. Each upgrade option is a paired contrast with QQ:

$$
\begin{aligned}
G &= Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},\\
H &= Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},\\
F &= Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(both)}.
\end{aligned}
$$

For a replacement, a *rescue* is a unit where the upgraded branch succeeds and QQ fails, and a *damage* is the reverse. *Turnover* is rescues plus damages; *net* is rescues minus damages. When the effect of the action swap differs between Q's and B's continuation, we call the outcome *continuation*

*dependent*. All contrasts measure effects on task success, not which action is correct. Appendix A.1 defines the symbols used in the appendix.

**Reference interventions.** A large turnover could mean that a decision needs higher precision, or only that it is sensitive to a change of model. We therefore rerun the design with Q′ in place of B: a second GPTQ-W4 checkpoint of the same model, calibrated on a re-drawn state and row sample from the same calibration trajectories, with the same Q episodes, tasks and nonce blocks and a protocol frozen before any outcome (q4 and q8 GPTQ-W4; GPTQ-W3 as a stress check). §5 adds a comparison at common states and a key-order change of Q's own action.

### 3.3 Making the counterfactuals auditable

**Replay acceptance.** Before any development outcome was analysed, an engineering gate on real divergence states passed every check: Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, every fork task re-ran the QQ no-op on an outcome-independent 10% hash sample of its states, and one mismatch would have quarantined the task's whole output; all 924 sampled development states passed, as did all sampled states of the other state sources and of the same-precision run (Appendix A). Exact replay is a tested property of this stack (greedy decoding, vLLM 0.29.0, batch-invariant kernels, deterministic compilation, dense BF16 kernels), not a general one (Yuan et al., 2025); determinism tests and untested conditions are in Appendix A.

### 3.4 Which states

**State sources.** A *first divergent state* is, for each task, block and configuration, the first decision at which B's and Q's actions differ in kind (tool call, text or stop), tool name, call count or order, or normalised arguments; two text replies that differ only in wording do not count. An *ordinary sampled state* is one decision per task and block, drawn by a hash fixed before generation, from Q's own trajectories or from BF16 trajectories shared by all configurations of a model. A *common state* is an ordinary sampled state at which several action sources are compared.

**Denominators and inference.** Rescues, damages, turnover and net use the *full pool* of task×block units; units without a first divergent state count as zero. $G$, $H$, $F$ and continuation dependence use the *divergent states* of a configuration; ordinary sampled states have one state per task and block. Within a configuration all branches share each state and all tasks of a block share its nonce, so contrasts are paired and identified; across configurations, which select their own divergences, comparisons are descriptive. Only complete blocks enter. The primary *fixed-pool* interval treats the task pool as fixed (block bootstrap with at least 10 blocks, else $t$ over block means); the *new-task* interval is a crossed task×block bootstrap ("crossed"). *Resolved* means that the named interval excludes zero; "not resolved" means that the reported interval or intervals include zero. Where a frozen protocol sets its own reading rule, we state it with the result.

## 4 Result 1: An Upgrade Has Two-Sided Consequences

The holdout is pre-registered; the upgrade-mode and pattern analyses are exploratory and post hoc.

**Rescues and damages are both large; the pooled net is not resolved.** On the $\tau^2$-bench development panel (85 tasks × 20 complete blocks), replacing a W4 agent's action at the first divergent state with B's, while Q continues, changes the task outcome in 15.5–20.9% of units (Figure 2a). Rescues (6.8–11.4 pp) and damages (7.5–10.1 pp) are of similar size, so the net lies between $-1.9$ and $+2.9$ pp. Two fixed-pool intervals exclude zero (negative for q4 GPTQ-W4, positive for q8 GPTQ-W4); no new-task interval does. The net is a local replacement effect, not an end-to-end comparison (Appendix A.2). Pooled over the four W4 configurations, the net is $+0.7$ pp (fixed pool $[-0.3, +1.6]$; new tasks $[-2.1, +3.4]$) and the turnover 18.6%.

**Which mode is evaluated decides which rescues are counted.** On divergent states no average gain is detected for upgrading this step, the rest, or both (W4 pool: $G = +0.5$, $H = +0.2$, $F = +0.7$ pp; all fixed-pool and new-task intervals cover zero; Appendix B.1). These average over divergent states only, so $G$ is not the full-pool net above. The averages hide states where single-mode evaluations disagree. In a *joint rescue* only BB succeeds (Figure 1); these are 4.0% of W4 divergent states (fixed-pool 95% [3.5, 4.5]; new-task [2.7, 5.3]), and testing only the current action ($G$) misses them. In a

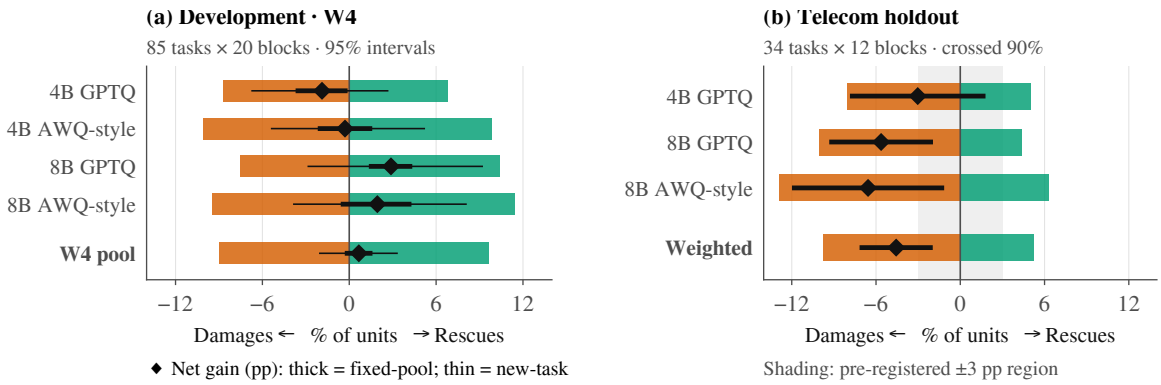



Figure 2: Replacing Q's action at the first divergent state with B's while Q continues: rescues (right) and damages (left) in % of task×block units; diamond: net (pp). (a) $\tau^2$-bench development panel, four W4 configurations and their pool (85 tasks × 20 blocks); thick: fixed-pool 95%, thin: new-task 95%. (b) Pre-registered telecom holdout (34 tasks × 12 blocks); bar: crossed 90% interval, the interval the frozen protocol reads; "weighted" is the estimate it reads, per-configuration rows are descriptive; shaded: the pre-registered small-effect region (±3 pp). All configurations: Appendix Tables 6 and 12.

*rescue undone*, B's action rescues under Q's continuation but fails under B's continuation; these are 4.8% (fixed-pool [4.1, 5.4]; new-task [3.2, 6.5]), and testing only the joint replacement ($F$) misses them. A label-permutation reference that keeps how many branches succeed produces both patterns at similar rates, so their frequency gives no evidence of enrichment that generalises to new tasks (exploratory; Appendix B.2). Whatever their cause, a gain measured for action replacement ($G$) cannot be reused as the gain of the joint replacement ($F$), or the reverse.

**On held-out telecom tasks the net estimate is negative, and the frozen reading is inconclusive.** On 34 unseen telecom tasks, under a protocol frozen before any holdout unit ran (12 fresh blocks, three W4 configurations), the replacement again rescued (5.2 pp) and damaged (9.8 pp). The primary net, −4.57 pp (crossed 90% [−7.17, −1.96]), lies below zero. The frozen readings had only positive-gain and small-effect categories, a design omission, so the protocol classifies the result as *inconclusive* (Figure 2b; Appendix C). Both pre-registered secondary checks passed: S1 (rescues and damages both positive) and S2 (cross-half repeatability of task fragility, $\rho = 0.588$, one-sided permutation $p = 0.0001$). All 135 QQ no-op replays selected by the fixed 10% hash rule, of 1202 divergent states, matched; this covers only the sampled states. The holdout changes domain as well as tasks (Appendix C.1).

**Pointers.** Outcome changes recur on the same tasks across nonce blocks (post hoc; this does not predict which new tasks are fragile; Appendix B.3). Exploratory follow-ups on label transfer and takeover duration are in Appendices G.1 and G.2; no main-text claim rests on them.

## 5 Result 2: Flips Alone Do Not Diagnose a Need for More Bits

**The question and its three designs.** If the consequences in §4 required a higher bit width, a substitution at the same bit width should produce markedly fewer of them. Table 1 lists three reference designs; the argument takes four steps, and Step 2 is a limit of the first design. Each is a reference intervention, compared with B's action only within its own population (status in Table 1), and never subtracted from B's turnover to obtain a precision-specific effect.

**Step 1: a higher bit width is not necessary for two-sided flips or continuation dependence.** Both also appear when only the calibration checkpoint is swapped. At W4 (six blocks, the longest complete prefix under the frozen rule), Q′'s action rescues 9.3 pp of units and damages 8.4 pp . Its turnover is 17.7%, against 16.8% for B on the same six blocks and two GPTQ-W4 configurations. The swap's effect depends on the continuation in 26.2% of its divergent states, against 25.0% for B. Each same-precision rate has a fixed-pool 95% lower bound above zero. A W3 stress pair also shows two-sided flips and continuation dependence when only the W3 checkpoint is swapped (turnover 17.3% for Q′ against 18.8% for B; Appendix D.1). Flip rate and continuation dependence alone therefore cannot diagnose a need for a higher bit width. We do

| Replacing action | Continues | Turnover | Rescues | Damages | Net | Paired difference [main interval] |
|---|---|---|---|---|---|---|
| **A. Second W4 checkpoint** | Own first divergent states; 1020 units, q4/q8, 85 tasks × 6 blocks | | | | | |
| B | Q | 16.8 | 8.0 | 8.7 | −0.7 | $\Delta M$ (B − Q′): −1.0 [−3.9, +1.9] |
| Q′ | Q | 17.7 | 9.3 | 8.4 | +0.9 | fixed-pool 95%; not resolved |
| **B. Common states** | 510 ordinary q8 states, 85 tasks × 6 blocks; all arms share the same states | | | | | |
| B | Q | 7.1 | 4.1 | 2.9 | +1.2 | $K$(Q): +1.0 [−4.2, +6.2] |
| Q′ | Q | 8.0 | 4.1 | 3.9 | +0.2 | two-look 97% $t$; not resolved |
| B | B | 9.0 | 5.3 | 3.7 | +1.6 | $K$(B): +0.8 [−2.2, +3.8] |
| Q′ | B | 8.6 | 4.7 | 3.9 | +0.8 | two-look 97% $t$; not resolved |
| **C. Argument key order** | 300 eligible q8 states, 41 tasks; one tool call with at least two argument keys | | | | | |
| B | Q | 9.0 | 3.3 | 5.7 | −2.3 | $\Delta M$ (reversed keys − B): −2.3 [−8.4, +2.9] |
| Q, keys reversed | Q | 6.7 | 3.3 | 3.3 | 0.0 | wider 95% bootstrap; not resolved |

Table 1: Reference interventions, in % of each group's units or states (net: pp). Rows are paired only within a group; references are not subtracted from B's turnover. Baseline: Q's action under the same continuation (QQ for Q, QB for B). A was frozen before outcomes. B's six-block extension and two-look rule were fixed after the first look; its readings require both $t$ and task-cluster bootstrap intervals. C is exploratory, with code frozen before outcomes. No equivalence is claimed. Full intervals and tests: Table 14.

not claim the turnovers are equal: neither pair is equivalent within ±3 pp.

**Step 2: because each design picks its own divergent states, this comparison cannot measure what B adds.** Each design forks at its own first divergence, so flips found by only one design cannot be read as caused by precision, nor flips found by both as not caused by it. The action-side net difference, −1.6 pp (B minus Q′; fixed pool [−6.2, +3.0], new tasks [−6.5, +3.6]), is not resolved and compares two selection-and-replacement protocols, not two actions at one state (Appendix D.1).

**Step 3: at common states, B's extra action value over Q′ is not resolved, and equivalence is not shown.** A comparison holds the states fixed: on 510 ordinary q8 GPTQ-W4 states (85 tasks × 6 blocks), the current action comes from B, Q or Q′, and B or Q continues (Table 1). What B's action adds beyond Q′'s at the same state and under the same continuation ($K$) is +1.0 pp under Q's continuation and +0.8 pp under B's. The design was pre-registered at three blocks, and that first look was unresolved; dated amendments then fixed an extension to six blocks and a two-look reading, before any extension run had produced output. Earlier studies had already produced outcomes at 26 of the 510 states; without them, the two-look reading is unchanged (Appendix D.2). Neither value is resolved under the protocol-designated two-look reading, and equivalence is not shown (Table 1; all intervals in Appendix D.2).

**Step 4: an outcome can change while the tool operation stays identical.** Steps 1–3 change which model writes the action; Step 4 changes only the token representation of Q's own action. At 300 first divergent states of q8 GPTQ-W4 where Q's action is one tool call with at least two argument keys (41 tasks; airline 187, retail 113), we reverse the order of the top-level keys. The reordered call was checked to parse and execute identically, so only the next tokens the agent reads change. This changes the outcome in 6.7% of these states ([0.8, 14.4]), against 9.0% [2.9, 18.2] for B's action on the same states. The difference, −2.3 pp [−8.4, +2.9], is not resolved, and we claim no equivalence. Some outcome changes therefore arise without any change to the tool operation. The result covers only these states and this one change, not free-text or other representation changes, and it is not a perturbation curve (Appendix D.3).

**What the four steps establish.** In these designs, two-sided flips and continuation dependence also appear when a same-bit-width checkpoint replaces Q's action (Step 1), and outcomes also change under one key-order change that leaves the tool operation identical (Step 4; exploratory). Their presence alone therefore does not show that a decision needs B. The comparison that isolates B's extra action value, at common q8 states, resolves neither an advantage nor equivalence (Step 3). Sensitivity to an action, the value higher precision adds at a state, and the gain of a given upgrade need separate measurements.

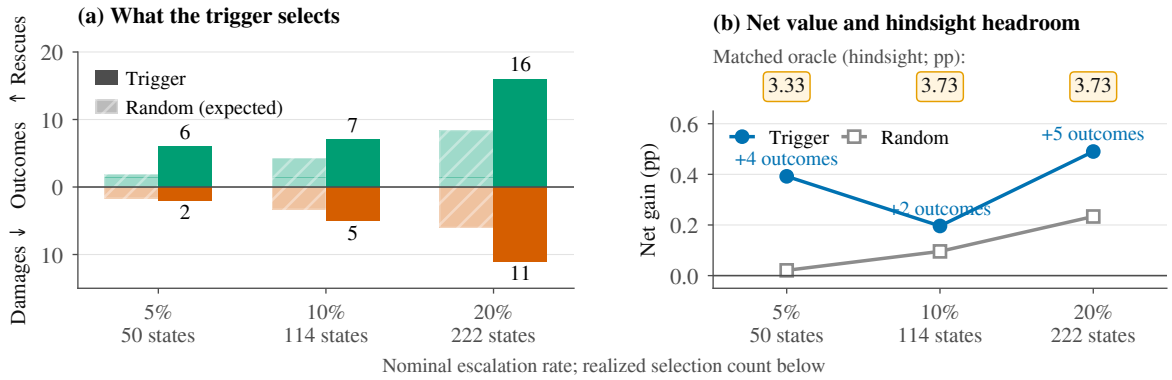



Figure 3: Frozen Q-only margin trigger on two later blocks of ordinary states from Q's own trajectories (six configurations; 1020 states; 38 rescues, 31 damages) at 5, 10 and 20% escalation. (a) Rescues (up) and damages (down) among the selected decisions: trigger (solid) and matched-count random selection (hatched, expected). (b) Net gain, (rescues − damages)/1020, in pp, trigger against random; labels give the trigger's net in outcomes. The numerical strip in (b) gives the matched-count hindsight oracle's net gain; it selects only rescues by construction. Nominal rates are shown with actual selected counts. Values and a post hoc KL reference: Appendix Table 15.

## 6 Result 3: A Frozen Sensitivity Trigger Does Not Reliably Select Value

**The frozen trigger enriches outcome-changing decisions at two of three rates.** A router needs a signal that picks net-positive upgrades, but $G$, $H$ and $F$ require running B, and a deployable rule sees only Q. On one development block, Q's token uncertainty separates states where B and Q act differently from states where they agree, but no Q-visible statistic reliably separates flips from non-flips among differing states (Appendix E.1). We then froze a Q-only trigger on that block, before any later state was scored: escalate one decision, meaning that B replaces only the current action and Q continues (the $G$ contrast), when the minimum Q margin over its action is low (margins predict cross-precision divergence; Du et al., 2026), with per-configuration thresholds targeting 5, 10 and 20% on the development block. These fixed thresholds do not impose exact quotas on later states; Figure 3 shows realised counts, and random selection matches those counts within each configuration. The margin score was chosen among three Q-only candidates (entropy, margin, mean log-probability) by its net gain over random on that block. On two later blocks of the development tasks (not the telecom holdout), at ordinary states from Q's own trajectories (six configurations: four W4, two KV8; 1020 states, 69 outcome-changing) it captured 8, 12 and 27 outcome-changing states, against 3.48, 7.52 and 14.30 expected at random ($p = .0182$,

| Scope check | Evidence in this setting | What it extends |
|---|---|---|
| BFCL multi-turn | q4/q8 W4, no simulated user; turnover 5.2–15.2%; pool net $+0.1$ $[-1.0, +1.2]$. q4 AWQ-style net $-3.8$ $[-6.5, -1.3]$. | Two-sided consequences; not a pooled net gain. |
| ALFWorld | q8 W4, 134 games, no simulated user; turnover 21.6–24.6%; pool net $-3.7$ $[-9.7, +1.9]$. | Two-sided consequences; not a pooled net gain. |
| Mistral-Small-3.2-24B | GPTQ/RTN-W4, ordinary BF16-sourced states, 83 tasks; GPTQ rescues 4.6 and damages 5.4 pp; no mode's net resolved. | Second family; no first-divergence or Q′ evidence. |
| Lower-bit stress | q8 RTN-W3, 255 first-divergence states, three blocks: $H = +15.3$, $F = +18.4$ [9.4, 28.2] pp (crossed 95%). | Positive takeover gains in a degraded setting; separate from W4. |
| QuaRot / SpinQuant | q8 W4A4, ordinary BF16-sourced states, three blocks. SpinQuant KL 0.165 vs 0.199; task-value contrast $+1.2$ $[-4.7, +7.1]$, unresolved. | Lower numerical error does not establish better task value. |
| KV-cache precision | KV8 at q4/q8 first divergences: turnover 6.5–8.1%. KIVI-2/4 at q8 ordinary states (two/three blocks): contrast against KV8 unresolved. | Two-sided turnover at KV8; no KIVI ordering. |

Table 2: Scope checks, with populations kept separate (Appendix F). BFCL and ALFWorld use one run per item, with post hoc item-bootstrap 95% intervals; the QuaRot/SpinQuant interval is crossed 95%.

.0613, .0002; Figure 3a). At ordinary states from BF16 trajectories (secondary), only the 20% rate enriches (Appendix E.1).

**Its net gain is not reliably above random.** The captured states were rescues and damages alike (6/2, 7/5 and 16/11). Net gains were $+4$, $+2$ and $+5$ outcomes, against random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided $p =$ .0345, .417, .262). No escalation rate was designated primary in advance, and the 5% result does not survive Holm correction over the three rates. A hindsight oracle with the same per-configuration counts nets $+34$, $+38$ and $+38$ outcomes ($+3.33$, $+3.73$ and $+3.73$ pp; Figure 3b). This headroom is positive by construction, because the oracle picks only rescues; the trigger selects 6, 7 and 16 of the oracle's 34, 38 and 38 rescues. A post hoc $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ score, which needs BF16, also captures turnover above chance on these states without a significant net gain (exploratory, post hoc; Appendix E.2); configuration-level KL analyses are in Appendix E.3.

**The negative result is specific to this rule.** Enriching outcome-changing decisions is not enough for reliable net gain (cf. Zhang, 2026). The later blocks reuse the development tasks, and only this simple frozen rule is tested. The result does not test learned or other uncertainty-based escalation (Li et al., 2026c; Piatrashyn et al., 2026; Lei and Huang, 2026) and does not refute sensitivity routing in general.

## 7 Discussion

**What an upgrade evaluation should report.** A study of precision routing or escalation should report the net task gain of the intervention it actually executes, as rescues and damages, together with that intervention's cost: the current action only, a continuation takeover, or both, under the continuation that runs afterwards. A gain measured for one of these is not the gain of another (§4). Enrichment of outcome-changing decisions validates a signal's sensitivity, not its value; our trigger result concerns one frozen Q-only rule, and learned routers such as DMR (Li et al., 2026c) and ReDAct (Piatrashyn et al., 2026) are untested (§6). Our experiments measure task value. They do not give a cost-optimal upgrade policy, and we measure no cost (simulated quantization; Limitations).

**How to read flips.** The two references show which consequences appear without more bits (§5). They are interventions on their own populations, not noise floors to subtract from B's turnover. Each branch measures a realised outcome under the fixed greedy execution stack; task/nonce intervals do not estimate a state's expectation over arbitrary simulators or semantic perturbations. One hypothesis, not an identified mechanism, is that a change in the history the agent reads propagates through the continuation. We tested one key-order change and specific checkpoint substitutions, not a perturbation curve, so the results do not say that any token change has these consequences.

**Precision is not shown to be irrelevant.** Two fixed-pool intervals exclude zero (uncorrected), the holdout estimate is negative (classified inconclusive by the frozen readings), and at common states equivalence of B's and Q′'s actions is not shown.

**Scope.** Table 2 lists the scope checks. At ordinary sampled states, most outcome changes occur where B's and Q's actions have the same tool structure and differ only in text; when a second user simulator replaced the first after a fixed history, we could not confirm their rescue and damage labels (Appendix F.1). All $\tau^2$-bench results use one greedy user simulator.

## 8 Conclusion

We evaluate a precision upgrade as a specified intervention at a restored state whose replay is audited: we count the rescues and damages of replacing the action, the continuation or both, and read them against a same-bit-width checkpoint and a representation change that keeps the tool operation identical. These references also change outcomes (two-sided for the same-bit-width checkpoint; observed in both directions, exploratory, for key order), and at common states B's extra action value over the same-bit-width checkpoint is not resolved; equivalence is not shown. Outcome sensitivity alone therefore does not show that more bits are needed. Upgrade evaluations should report the rescues and damages of the executed intervention under the intended continuation, read them against reference interventions, and validate selection rules by net value, not flip enrichment.

## Limitations

**What we do not claim.** We do not claim success equivalence between BF16 and W4 agents (equivalence tests at $\pm 3$ pp on new-task intervals fail), equivalence of BF16 and a second W4 checkpoint as replacements, a W$\times$KV interaction, a distinct action–continuation interaction mechanism, or that outcomes are mainly determined by the action. We do not present static precision restoration as a method (Appendices H and H.1). $G$, $H$ and $F$ are oracle quantities, not the value of a deployable policy (§6). The outcome classes and the label-permutation reference (Appendix B.2) are exploratory and post hoc, with headline shares from the retail and airline development panel. The action-over-continuation gap comes only from the 13.8% of W4 divergent states with exactly two successful arms. Label transfer between modes and takeover duration are exploratory follow-ups (Appendix G); no main-text claim rests on them.

**Two model families, limited coverage.** Most crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. The second family, Mistral-Small-3.2-24B (GPTQ-W4, RTN-W4), was forked only at ordinary BF16-sourced $\tau^2$-bench states of an 83-task conditional population (Appendix F.3); it has no first-divergent-state forks, no same-precision reference and no outcome-class analysis. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix A.3). A time-boxed Llama-3.1-8B-Instruct check (GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no evidence for that family.

**Simulated quantization.** Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

**One user simulator; two small controls.** All $\tau^2$-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The first-divergent-state control has 20 states per configuration, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw: the replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the ordinary-state control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation; the first-divergent-state control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel (six blocks) and GPTQ-W3 on three blocks, and swaps one calibration sample for another, re-drawn from the same calibration trajectories; it does not cover checkpoints calibrated on independent datasets. Six is the longest complete common prefix under the frozen rule, because the seventh block of the same-precision q8 design is incomplete. It is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8, RTN or rotated KV4.

**Small confirmation and partial state sets.** The pre-registered holdout covers 34 telecom tasks and three configurations. Its pre-registered readings had only positive-gain and small-effect categories; omitting a negative-effect category was a design omission, and it leaves the protocol able to classify the negative estimate only as inconclusive. It applies only to telecom, these three configurations and this simulator. A post hoc audit found that all 135 QQ no-op replays selected by the fixed 10% hash rule, of 1202 divergent states, matched; it covers only those sampled states (Appendix C). Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout configurations show no detectable net effect (Appendix C.1), so the telecom domain alone does not produce a detectable drop. But their 6-block intervals are wide and q4 has no telecom development forks, so the holdout drop may reflect heterogeneity among its 34 tasks as much as a domain effect. First-divergent-state forks intervene only at the first tool-structure divergence per task and

block. Ordinary-state forks sample one state per task and block; their main analyses use three blocks, so their block intervals rest on two degrees of freedom, and the later panels use two to six blocks. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

## Ethics Statement

This work involves no crowdworkers, no human annotation and no personal data. Where a user is present ($\tau^2$-bench), it is simulated by an LLM; tasks are public benchmark tasks, and all agents are openly released Qwen3, Mistral-Small-3.2 and (gate check only) Llama-3.1 checkpoints, or simulated-quantized versions of them. Outcomes are the benchmarks' automatic success checks, and action categories are assigned by fixed rules; reported values map to committed analysis outputs (Appendix H.1). The main risk we see is overreading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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## A Setup, Replay Acceptance and Notation (supports §3)

This appendix supports §3: it gives the splits, how a state is restored and forked, which replay checks passed on this serving stack, and the internal name behind each main-text term.

**Splits.** The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from $\tau^2$-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

**Success endpoints.** For retail and airline we use the final-database match with normal termination. This is not $\tau^2$-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

**Fork mechanics.** The forced action is returned by a fork agent for the one step that follows $h_t$. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. $\tau^2$-bench resets step and error counters when it starts from a history. The fork restores them to the values used at $t$, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

**Replay acceptance details.** In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Besides the 924 sampled development states, 328/328 sampled Bank U states passed the QQ no-op, and the Bank U gate passed a further 135/135 no-ops (a different set from the holdout's 135 QQ no-ops, all of which also passed; Appendix C). Serving in separate processes closes the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.

**Determinism of the serving stack.** Closed-loop determinism pairs, the same configuration on two nodes at different times, reproduced whole episodes token for token for dense BF16 paths at tensor parallelism 1 and 2 with 1 to 16 concurrent requests (e.g. 257/257 and 1520/1520 episodes). Nine arms that two studies ran independently at the same states were byte-identical. We did not test exactness under KV-cache preemption, or on other models, GPUs or vLLM versions. A deployed AWQ kernel path with fp16 dequantization was not replay-exact, and no reported outcome uses it.

**User simulator.** The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

### A.1 Internal names and notation

**Glossary.** The main text uses plain terms; this supplement also uses the internal names of the analysis code. A *first divergent state* is a *Bank P* state. An *ordinary sampled state* is a *Bank U* state; its sub-banks are drawn from Q's own trajectories (U1) or from trajectories shared by all configurations of a model (U2B: BF16 trajectories; U2Q: Q trajectories). A *common state* is an ordinary state of the common-state design (pre-registered at three blocks, extended to six by amendment), at which B, Q and Q′ each supply the current action. A *cell* is a configuration, named by model (q4: Qwen3-4B; q8: Qwen3-8B; q14: Qwen3-14B; m24: Mistral-Small-3.2-24B) and quantization setting. Nonce blocks carry ids such as fNN (development) and qN (stress and W3 panels); the holdout blocks have their own ids. Branches are written by action source and continuing model (BB, BQ, QB, QQ; KQ is the key-order branch, Appendix D.3). Symbols: $r$ and $d$ are the rescue and damage rates of the action swap, $M = r + d$ is the turnover and $G_{\text{all}} = r - d$ the net gain on the full pool; $\Delta_{\text{Q}}$, $\Delta_{\text{B}}$ and the off-diagonal share are defined below; $I = \Delta_{\text{Q}} - \Delta_{\text{B}}$ is the interaction of action and continuation (Table 7); $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ compares methods on shared ordinary states (Appendix F.5); $K(c)$ is B's action value beyond Q′'s at a common state under continuation $c$ (Appendix D.2); $n_{\text{blk}}$ is the number of complete blocks.

**Action swap, rescues and damages.** The paired effect of the action swap under each continuation is $\Delta_{\text{Q}} = Y_{\text{BQ}} - Y_{\text{QQ}}$ $(= G)$ and $\Delta_{\text{B}} = Y_{\text{BB}} - Y_{\text{QB}}$, each in $\{-1, 0, +1\}$. When $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$ ("off-diagonal"), the swap's value depends on the continuation. For the action swap, a *rescue* is $(Y_{\text{BQ}}, Y_{\text{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates $r$ and $d$, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

**Same-precision notation.** $M'$, $r'$, $d'$ and the off-diagonal share are defined likewise with Q′ for B.

**Which states.** *Bank P*: for each task, block and cell, the first decision at which $a_{\mathrm{B}}$ and $a_{\mathrm{Q}}$ differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks (up to six for U2B in §5–§7), from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

**Denominators and inference.** We keep three denominators separate. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; $r$, $d$, $M$ and $G_{\mathrm{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; $G$, $H$, $F$ and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. With $n_{\mathrm{blk}}$ complete blocks (only complete blocks enter), the primary *fixed-pool* interval treats the task pool as fixed (block bootstrap if $n_{\mathrm{blk}} \geq 10$, else block-$t$, a $t$ interval over block means); the *new-task* interval is a crossed task×block bootstrap ("crossed"). We call a contrast *resolved* under an interval when that interval excludes zero and name the interval; "not resolved" means the reported interval or intervals include zero. Where a frozen protocol sets its own reading rule, we state it with the result.

### A.2 Separating local value from end-to-end success

The replacement gains $G_{\mathrm{all}}$, $G$, $H$ and $F$ are not end-to-end comparisons. Table 3 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

### A.3 Coverage Matrix

Table 4 lists the completed crossed-intervention coverage behind every Qwen3 Bank P and Bank U result; the coverage of the U2B panels (§6, §7) is in Appendix F.5. It counts tasks, states and complete blocks after the strict admission rule. Four of the five stress cells turn over 18.4–20.4% of units with three blocks; we keep them separate from W4, and no W2 crossed-intervention panel is reported.

### A.4 Quantization configurations

Table 5 gives the quantization settings of every checkpoint configuration, read from the checkpoint metadata and the quantization code. Not recorded: the AWQ-style row-sample size (4096 is the code default) and the KV-cache scale dtype (the hook computes in fp32); Mistral has no second calibration draw.

## B Two-Sided Consequences in Full (supports §4)

This appendix supports §4: replacing a W4 agent's action produces both rescues and damages, and a gain measured for one upgrade mode cannot be reused as the gain of another.

Table 6 gives every cell of Figure 2a, including the stress cells, with new-task intervals. Table 7 gives $G$, $H$, $F$ and their paired differences for every cell, with $F - G$, $F - H$ and the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ summaries.

### B.1 New-task intervals for the four-branch estimands

Table 8 gives all new-task intervals corresponding to Table 7.

### B.2 Outcome Classes and a Label-Permutation Reference

Everything in this section is exploratory and post hoc. It uses the four outcomes of each divergent state in the order BB, BQ, QB, QQ; $k$ is the number of successful arms.

**Five classes.** The 16 patterns fall into five classes: unchanged (`0000`, `1111`); action only (`1100`, `0011`: the outcome follows the action under both continuations); continuation only (`1010`, `0101`); one arm differs from the other three ($k$ odd); strict reversal (`1001`, `0110`). Changing the action changes the outcome under at least one continuation exactly in the action-only, one-arm and reversal classes, and changing the continuation exactly in the continuation-only, one-arm and reversal classes. Both change it exactly on the off-diagonal states, so the two shares overlap and must not be added, and their gap equals action only minus continuation only: it comes from $k = 2$ states alone and restates the $k = 2$ arrangement below on all divergent states. Two named patterns follow the upgrade modes: joint rescue (`1000`: only the joint upgrade succeeds) and rescue undone ($Y_{\mathrm{BQ}} = 1$, $Y_{\mathrm{QQ}} = 0$,

| Cell | Q / BF16 (%) | Q−BF16 [95% CI] | NI 3 pp | TOST ±3 pp |
|---|---|---|---|---|
| q4 GPTQ-W4 | 40.0 / 35.8 | +4.2 [−2.4, +10.5] | pass | fail |
| q4 AWQ-style W4 | 36.8 / 35.8 | +0.9 [−5.8, +7.5] | fail | fail |
| q8 GPTQ-W4 | 30.1 / 30.4 | −0.2 [−7.7, +6.8] | fail | fail |
| q8 AWQ-style W4 | 31.5 / 30.4 | +1.2 [−5.3, +7.6] | fail | fail |
| pool, all four | | +1.5 [−2.7, +5.6] | pass | fail |

Table 3: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

| Cell | $\tau^2$-bench retail (Bank P) | $\tau^2$-bench airline (Bank P) | telecom dev (Bank P) | Bank U (retail / airline) | BFCL / ALFWorld |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | 58 t · 727 st · 20/20 b | 27 t · 375 st · 20/20 b | holdout only | 3 b · 393 / 177 rows | 788 items · 573 st / – |
| q4 AWQ-style W4 | 58 t · 986 st · 20/20 b | 27 t · 453 st · 20/20 b | – | 3 b · 393 / 177 rows | 788 items · 642 st / – |
| q4 KV8 | 58 t · 412 st · 22/22 b | 27 t · 181 st · 22/22 b | – | 3 b · 393 / 177 rows | – / – |
| q8 GPTQ-W4 | 58 t · 1017 st · 20/20 b | 27 t · 456 st · 20/20 b | 80 t · 477 st · 6/6 b; holdout | 3 b · 393 / 177 rows | 788 · 533 st / 134 · 115 st |
| q8 AWQ-style W4 | 58 t · 1087 st · 20/20 b | 27 t · 475 st · 20/20 b | 80 t · 480 st · 6/6 b; holdout | 3 b · 393 / 177 rows | 788 · 548 st / 134 · 119 st |
| q8 KV8 | 58 t · 585 st · 22/22 b | 27 t · 270 st · 22/22 b | – | 3 b · 393 / 177 rows | – / – |
| q4 GPTQ-W3 | 58 t · 162 st · 3/3 b | 27 t · 70 st · 3/3 b | – | – | 591 · 540 st† / – |
| q4 KV4-rot | 58 t · 164 st · 3/3 b | 27 t · 69 st · 3/3 b | – | – | – / – |
| q8 GPTQ-W3 | 58 t · 166 st · 3/3 b | 27 t · 76 st · 3/3 b | – | – | 788 · 696 st / 134 · 130 st |
| q8 KV4-rot | 58 t · 165 st · 3/3 b | 27 t · 74 st · 3/3 b | – | – | – / – |
| q8 RTN-W3 | 58 t · 174 st · 3/3 b | 27 t · 81 st · 3/3 b | – | – | – / – |
| q4 / q8 KV4 (unrotated) | – | – | – | – | 788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st |
| q14 GPTQ-W4 | – | – | – | – | 788 · 471 st / – |

Table 4: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix F.2). "holdout": the cell is in the telecom holdout (Appendix C). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix C.1) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix F.4). The end-to-end q4 AWQ-W3 run is incomplete (two airline episodes hit 600 s serving timeouts), and so are the q4 RTN-W2 stress runs (context-window overflows of the degenerate model and a truncated result file); neither is reported.

$Y_{\mathrm{BB}} = 0$: B's action rescues under Q's continuation and B's takeover of the continuation undoes it). Table 9 gives the shares. The action-side gap holds in the $\tau^2$-bench retail/airline W4 and KV8 pools, in the W4 same-precision fork and in the BFCL W4 pool. It is near zero on the telecom development tasks (+0.1 [−2.1, +2.4] pp, 6-block $t$) and in the q4 GPTQ-W3 and KV4-rot stress cells, and it is not resolved in the ALFWorld W4 pool (+3.4 [−1.8, +8.4]) or the W3 same-precision pair (+1.9 [−2.9, +6.7]). Its point estimate is negative in the q8 RTN-W3 stress cell (−7.8; 3-block $t$ [−18.1, +2.4], new-task [−15.3, −1.2]) and, not shown, in ALFWorld q8 GPTQ-W3 (−19.2 [−27.3, −10.8]). In the ALFWorld and BFCL KV4 cells (not shown) Q almost never succeeds, so continuation-only states dominate there mechanically.

**W4 pool details.** In the $\tau^2$-bench W4 pool, the gap between the action and continuation shares is

| Config. (models) | Method, weights | Group, grid | Act-order | Calibration | KV cache / activations / rotation |
|---|---|---|---|---|---|
| GPTQ-W4 Q (q4, q8) | GPTQ, W4; damp 0.01, block 128, true-sequential | 128, asym. | yes | draw c1: 57 BF16 agent-trajectory states (39 retail, 18 airline calibration tasks, disjoint from the 85 dev tasks), one state per task, full context (median 6041 / 6310 tok.); 64,000 activation rows (1122–1123 per state); draw seed 20260729 | KV BF16, act. BF16, no rotation |
| GPTQ-W4 Q′ (q4, q8) | as Q | 128, asym. | yes | draw c2, seed 20260730: same 57 tasks and trajectories, re-drawn decision state and rows (5/57 states shared with c1); not disjoint | as Q |
| AWQ-style W4 (q4, q8) | AWQ-style: per-linear scale $\lvert\bar{x}\rvert^{\alpha}$, 20-point $\alpha$ grid, scale folded into weight, then RTN; no clip search | 128, asym. | n/a (GPTQ only) | draw c1 as GPTQ-W4 Q; $\alpha$ fit on a 4096-row sample (code default; per-run value not recorded) | as Q |
| GPTQ-W3$^{s}$ Q / Q′ (q4, q8) | GPTQ, W3 | 128, asym. | yes | c1 / c2 as W4 | as Q |
| RTN-W3$^{s}$ (q4, q8) | RTN, W3 | 128, asym. | n/a | none | as Q |
| KV8 (q4, q8) | BF16 weights | – | – | none | INT8 K and V, asym. min–max per (token, KV head) over head_dim, no clip, no rotation; fake quant in fp32 |
| KV4-rot$^{s}$ (q4, q8) | BF16 weights | – | – | none | INT4 K and V, same grid as KV8; online normalized Hadamard on head_dim: q and k rotated, V rotated and rotated back |
| QuaRot-W4A4 (q8) | random-Hadamard rotation (seed 0) + GPTQ W4, MSE clip search | per out-channel, sym. | no | WikiText-2 train, 128 × 2048 tok., seed 0 | A4: sym. per-token dynamic, clip 0.9, inputs of qkv, o, gate/up, down, online Hadamard (o, down); KV16, per-head q/k Hadamard after RoPE |
| SpinQuant-W4A4 (q8) | learned $R_1$, per-layer $R_2$ (100 steps, lr 1.5, A4 in training) + GPTQ W4 as QuaRot | per out-channel, sym. | no | weights and rotations: WikiText-2, 128 × 2048, seed 0 | as QuaRot-W4A4 |
| KIVI-2 / KIVI-4 (q8) | BF16 weights | – | – | none | K per channel, V per token, group 32, asym.; last 128 tokens full precision |
| Mistral GPTQ-W4 / RTN-W4 (m24) | GPTQ W4 / RTN W4 | 128, asym. | yes / n/a | GPTQ: draw c1, 57 states, 64,000 rows, seed 20260729 (no c2 draw) | as Q |

Table 5: Quantization configuration of every checkpoint configuration, from the checkpoint metadata. AWQ-style W4 is AWQ-style (scale folded into the weight, then RTN, no clip search). Q′ re-draws the decision state and activation rows from the same 57 calibration trajectories as Q. The W4A4 configurations keep a 16-bit KV cache and calibrate on WikiText-2, so a QuaRot-minus-GPTQ contrast also changes the calibration source. One q8 calibration context (c1) is marked truncated. All checkpoints are fake-quantized and stored in BF16; embeddings, `lm_head` and norms stay BF16. $^{s}$stress cell.

+5.2 pp (new-task [+3.1, +7.7]): action effects are somewhat more common, and the continuation changes the outcome mostly where the action matters too. Of the 26.9% off-diagonal states, 23.8% have one arm differing from the other three and 3.2% are strict reversals; states where the swap helps more under Q ($I > 0$, 13.6%) and under B ($I < 0$, 13.4%) are about equally frequent, so the

| Cell | units | div. | $r$ | $d$ | $M$ | $G_{\text{all}}$ [fixed pool] | [new tasks] |
|---|---|---|---|---|---|---|---|
| q4 GPTQ-W4 | 85×20 | 1102 | 6.8 | 8.7 | 15.5 | $-1.9$ [$-3.7$, $-0.1$] | [$-6.8$, $+2.7$] |
| q4 AWQ-style W4 | 85×20 | 1439 | 9.8 | 10.1 | 19.9 | $-0.3$ [$-2.2$, $+1.6$] | [$-5.4$, $+5.2$] |
| q8 GPTQ-W4 | 85×20 | 1473 | 10.4 | 7.5 | 17.9 | $+2.9$ [$+1.4$, $+4.4$] | [$-2.9$, $+9.2$] |
| q8 AWQ-style W4 | 85×20 | 1562 | 11.4 | 9.5 | 20.9 | $+1.9$ [$-0.6$, $+4.3$] | [$-3.9$, $+8.1$] |
| W4 pool | 85×20 | | 9.6 | 9.0 | 18.6 | $+0.7$ [$-0.3$, $+1.6$] | [$-2.1$, $+3.4$] |
| q4 KV8 | 85×22 | 593 | 3.1 | 3.4 | 6.5 | $-0.3$ [$-1.7$, $+1.0$] | [$-3.0$, $+2.3$] |
| q8 KV8 | 85×22 | 855 | 4.3 | 3.7 | 8.1 | $+0.6$ [$-0.4$, $+1.6$] | [$-1.9$, $+3.0$] |
| q4 GPTQ-W3$^s$ | 85×3 | 232 | 9.0 | 10.2 | 19.2 | $-1.2$ [$-7.0$, $+4.7$] | [$-8.6$, $+5.9$] |
| q4 KV4-rot$^s$ | 85×3 | 233 | 11.8 | 7.1 | 18.8 | $+4.7$ [$-4.1$, $+13.5$] | [$-3.1$, $+12.9$] |
| q8 GPTQ-W3$^s$ | 85×3 | 242 | 10.2 | 8.2 | 18.4 | $+2.0$ [$-12.5$, $+16.4$] | [$-6.3$, $+11.8$] |
| q8 KV4-rot$^s$ | 85×3 | 239 | 11.8 | 8.6 | 20.4 | $+3.1$ [$-14.5$, $+20.8$] | [$-5.9$, $+12.9$] |
| q8 RTN-W3$^s$ | 85×3 | 255 | 3.1 | 3.5 | 6.7 | $-0.4$ [$-3.8$, $+3.0$] | [$-4.7$, $+3.9$] |

Table 6: Replacing Q's first tool-structure-divergent action while Q continues, $\tau^2$-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix A.2). Fixed pool: block bootstrap (≥10 blocks) or $t$ over block means; new tasks: crossed task×block bootstrap. $^s$stress cell, 3 blocks.

mean interaction is $+0.0$ [$-1.9$, $+2.0$] pp. With fixed-pool, then new-task 95% intervals, changing the action changes the outcome under at least one continuation in 34.9% ([33.6, 36.0]; [28.8, 41.0]) of W4 divergent states, and changing the continuation controller does so under at least one action in 29.6% ([28.5, 30.7]; [24.2, 35.3]). The two sets overlap in the 26.9% ([25.7, 28.1]; [22.1, 32.0]) where the swap's effect depends on the continuation, so the shares must not be added; all four arms agree in 62.4%.

**A label-permutation reference.** Given $k$, assigning the four outcomes to the four arms uniformly at random gives the off-diagonal share $E_0 = P(k \text{ odd}) + P(k = 2)/3$. In every cell the observed share minus $E_0$ equals $P(k = 2)\,[P(\text{reversal} \mid k = 2) - 1/3]$, so it can depart from $E_0$ only through strict reversals. In the W4 pool the share is 26.9% against $E_0 = 28.4\%$ ($-1.4$ [$-1.8$, $-1.0$] pp, fixed pool, block bootstrap over 20 blocks; KV8: 15.3 against 18.9%), because reversals are 23.6% [19.2, 28.3] (task bootstrap) of $k = 2$ states instead of one third. This shows only that strict reversals are rarer than under random labelling. It does not show that the continuation controller is irrelevant, which would require $Y_{\text{BB}} = Y_{\text{BQ}}$ and $Y_{\text{QB}} = Y_{\text{QQ}}$, or that the interaction is zero: the reference itself allows $I \neq 0$ in most arrangements. Conversely, an off-diagonal state does not show a distinct interaction mechanism: if an arm succeeds when an action score plus a continuation score exceeds a threshold, exactly additive scores can make only BB succeed (`1000`), although they never produce a strict reversal.

**Joint rescue and rescue undone against the reference.** The same reference, applied to the two named patterns, keeps $k$ for every state and permutes the four outcome labels uniformly: `1000` then has probability $1/4$ at $k = 1$, and rescue undone (`0100` or `0110`) has $1/4$ at $k = 1$ plus $1/6$ at $k = 2$, so the reference shares are $P(k{=}1)/4$ and $P(k{=}1)/4 + P(k{=}2)/6$, computed on the same states and bootstrap draws as the observed shares (Table 10). In the $\tau^2$-bench W4 pool, joint rescue exceeds the reference by $+0.47$ pp on the fixed task pool ([$+0.06$, $+0.87$]), but the new-task interval spans zero ([$-0.33$, $+1.37$]); rescue undone is less frequent than the reference ($-0.99$ [$-1.58$, $-0.42$]; new tasks [$-2.19$, $+0.27$]). The excess of joint rescue is concentrated in the q8 cells, and no group shows a new-task excess of rescue undone. Because the reference keeps $k$ and does not model execution, a share near it says that the arrangement is about as common as a random arrangement of the same number of successes, not that the outcomes are noise or that the continuation is irrelevant.

**The $k = 2$ arrangement.** At $k = 2$ the two successes share the action in 56.4% of W4 states and 80.3% of KV8 states, against one third under random labelling. A statement that outcomes are mainly determined by the action holds only within this subset, which is 13.8% (W4) and 14.5% (KV8) of divergent states.

**A. Gains and action–continuation comparison**

| Cell | states | $G$ (this step) | $H$ (next step on) | $F$ (both) | $G - H$ |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | 1102 | $-2.9$ [$-5.7$, $-0.2$] | $+0.7$ [$-1.6$, $+3.0$] | $-2.5$ [$-5.3$, $+0.2$] | $-3.6$ [$-6.6$, $-0.6$] |
| q4 AWQ-style W4 | 1439 | $-0.3$ [$-2.6$, $+1.9$] | $+0.9$ [$-1.1$, $+2.9$] | $-0.8$ [$-3.3$, $+1.6$] | $-1.3$ [$-3.0$, $+0.6$] |
| q8 GPTQ-W4 | 1473 | $+3.3$ [$+1.6$, $+5.0$] | $+0.4$ [$-1.2$, $+1.9$] | $+5.4$ [$+3.5$, $+7.2$] | $+2.9$ [$+1.3$, $+4.7$] |
| q8 AWQ-style W4 | 1562 | $+2.1$ [$-0.6$, $+4.7$] | $-1.3$ [$-3.5$, $+0.9$] | $+0.6$ [$-1.5$, $+2.7$] | $+3.4$ [$+1.5$, $+5.2$] |
| W4 pool | 5576 | $+0.5$ [$-0.7$, $+1.7$] | $+0.2$ [$-1.3$, $+1.6$] | $+0.7$ [$-0.6$, $+2.1$] | $+0.4$ [$-0.8$, $+1.7$] |
| q4 KV8 | 593 | $-1.0$ [$-5.3$, $+3.2$] | $-2.5$ [$-4.5$, $-0.5$] | $-1.2$ [$-4.7$, $+2.3$] | $+1.5$ [$-3.1$, $+6.0$] |
| q8 KV8 | 855 | $+1.3$ [$-0.9$, $+3.4$] | $-0.4$ [$-2.1$, $+1.5$] | $+2.5$ [$+0.0$, $+5.1$] | $+1.6$ [$-0.6$, $+3.9$] |
| KV8 pool | 1448 | $+0.1$ [$-2.8$, $+3.0$] | $-1.4$ [$-2.7$, $-0.1$] | $+0.6$ [$-1.8$, $+3.1$] | $+1.6$ [$-1.2$, $+4.1$] |
| q4 GPTQ-W3$^s$ | 232 | $-1.3$ [$-7.7$, $+5.0$] | $+4.3$ [$-10.7$, $+19.4$] | $+1.3$ [$-1.9$, $+4.4$] | $-5.6$ [$-15.0$, $+3.7$] |
| q4 KV4-rot$^s$ | 233 | $+5.2$ [$-4.5$, $+14.9$] | $+9.4$ [$+5.1$, $+13.7$] | $+12.4$ [$+11.0$, $+13.9$] | $-4.3$ [$-18.3$, $+9.8$] |
| q8 GPTQ-W3$^s$ | 242 | $+2.1$ [$-13.3$, $+17.2$] | $+0.4$ [$-7.4$, $+8.2$] | $+2.1$ [$-7.4$, $+11.4$] | $+1.7$ [$-9.1$, $+12.3$] |
| q8 KV4-rot$^s$ | 239 | $+3.3$ [$-15.4$, $+21.9$] | $-0.8$ [$-18.2$, $+16.3$] | $+4.6$ [$-16.0$, $+25.0$] | $+4.2$ [$+2.5$, $+5.8$] |
| q8 RTN-W3$^s$ | 255 | $-0.4$ [$-3.8$, $+3.0$] | $+15.3$ [$+0.7$, $+29.9$] | $+18.4$ [$+16.7$, $+20.1$] | $-15.7$ [$-27.5$, $-3.9$] |

**B. Joint-upgrade contrasts and outcome patterns**

| Cell | $F - G$ | $F - H$ ($= \Delta_{\mathrm{B}}$) | off-diag. % | rev. % | mean $I$ |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | $+0.5$ [$-1.2$, $+2.2$] | $-3.2$ [$-6.4$, $+0.2$] | 25.0 | 2.6 | $+0.3$ |
| q4 AWQ-style W4 | $-0.4$ [$-2.4$, $+1.5$] | $-1.7$ [$-3.6$, $+0.4$] | 29.3 | 3.1 | $+1.3$ |
| q8 GPTQ-W4 | $+2.1$ [$+0.2$, $+4.0$] | $+5.0$ [$+3.2$, $+6.8$] | 22.9 | 2.7 | $-1.7$ |
| q8 AWQ-style W4 | $-1.5$ [$-4.2$, $+0.8$] | $+1.9$ [$-0.1$, $+3.8$] | 30.5 | 4.2 | $+0.3$ |
| W4 pool | $+0.2$ [$-1.0$, $+1.2$] | $+0.5$ [$-1.0$, $+2.1$] | 26.9 | 3.2 | $+0.0$ |
| q4 KV8 | $-0.2$ [$-3.5$, $+3.7$] | $+1.3$ [$-2.5$, $+4.9$] | 15.7 | 1.2 | $-2.4$ |
| q8 KV8 | $+1.2$ [$-0.5$, $+3.2$] | $+2.8$ [$+0.3$, $+5.5$] | 14.9 | 1.2 | $-1.5$ |
| KV8 pool | $+0.5$ [$-1.6$, $+2.8$] | $+2.1$ [$-0.2$, $+4.2$] | 15.3 | 1.2 | $-1.9$ |
| q4 GPTQ-W3$^s$ | $+2.6$ [$-0.6$, $+5.8$] | $-3.0$ [$-15.1$, $+9.1$] | 31.5 | 3.9 | $+1.7$ |
| q4 KV4-rot$^s$ | $+7.3$ [$-2.0$, $+16.5$] | $+3.0$ [$-1.9$, $+8.0$] | 33.0 | 1.7 | $+2.1$ |
| q8 GPTQ-W3$^s$ | $+0.0$ [$-11.0$, $+11.1$] | $+1.7$ [$-0.1$, $+3.4$] | 23.1 | 4.5 | $+0.4$ |
| q8 KV4-rot$^s$ | $+1.3$ [$-1.8$, $+4.3$] | $+5.4$ [$+0.8$, $+10.0$] | 31.0 | 4.6 | $-2.1$ |
| q8 RTN-W3$^s$ | $+18.8$ [$+13.8$, $+23.9$] | $+3.1$ [$-13.0$, $+19.2$] | 29.0 | 0.4 | $-3.5$ |

Table 7: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$, $H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}}$, $F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}}$. Off-diag.: share of states with $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$; rev.: strict reversals, $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$; mean $I$: mean action–continuation interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}} \in \{-2, \ldots, +2\}$. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; $t$ over 3 blocks for $^s$stress cells, which cannot be pooled). New-task intervals are in Table 8. Within a row, contrasts are paired and identified; between rows, descriptive. Intervals are not adjusted for multiplicity: the four core cells' $G - H$ form a Holm family ($m = 4$; q4 GPTQ-W4 Holm $p = 0.069$), their new-task intervals all cover zero, and $G - H > 0$ ranks two modes, not either against QQ.

### B.3 Task fragility

All analyses in this subsection are exploratory and post hoc.

**The consequences recur on the same tasks.** We split the 20 nonce blocks into two halves by an outcome-independent hash of the block name, ranked tasks by turnover on one half and measured the other half (Appendix Table 11). The 17 most fragile tasks (top 20%) carry 41.0% (crossed [26.6, 55.9]) and 42.3% ([27.0, 57.5]) of the held-out turnover in the two directions, against 20% if fragility did not persist. Their turnover is 38.7%, while the other 68 tasks still turn over 13.7%. Both groups have rescues and damages; neither group's net gain is resolved under the crossed interval. This replicates fragility on other nonces of the same tasks; it does not predict which new tasks are fragile.

**Split-half fragility (Table 11).** The halves are assigned by an outcome-independent hash of the block name (sha256 of a fixed prefix and the block name, mod 2), written to a file before this post hoc analysis loaded outcomes; fragility is a task's mean turnover over the four W4 cells and the defining

**A. Upgrade gains**

| Cell | $G$ | $H$ | $F$ |
|---|---|---|---|
| q4 GPTQ-W4 | $[-10.3, +4.2]$ | $[-4.0, +6.0]$ | $[-9.9, +4.7]$ |
| q4 AWQ-style W4 | $[-6.5, +6.1]$ | $[-3.7, +5.5]$ | $[-7.4, +5.8]$ |
| q8 GPTQ-W4 | $[-3.3, +10.5]$ | $[-3.8, +4.9]$ | $[-2.0, +13.0]$ |
| q8 AWQ-style W4 | $[-4.1, +8.7]$ | $[-7.1, +4.6]$ | $[-6.2, +7.3]$ |
| W4 pool | $[-2.9, +3.8]$ | $[-2.7, +2.8]$ | $[-3.1, +4.4]$ |
| q4 KV8 | $[-9.4, +7.3]$ | $[-6.4, +1.1]$ | $[-9.3, +7.7]$ |
| q8 KV8 | $[-4.0, +6.9]$ | $[-3.5, +2.9]$ | $[-3.5, +8.1]$ |
| KV8 pool | $[-5.3, +5.6]$ | $[-3.7, +0.9]$ | $[-4.5, +5.8]$ |
| q4 GPTQ-W3$^s$ | $[-9.7, +6.6]$ | $[-6.6, +14.5]$ | $[-6.8, +9.4]$ |
| q4 KV4-rot$^s$ | $[-3.4, +14.2]$ | $[+2.1, +17.0]$ | $[+3.0, +21.6]$ |
| q8 GPTQ-W3$^s$ | $[-6.7, +12.1]$ | $[-7.2, +8.1]$ | $[-6.6, +11.8]$ |
| q8 KV4-rot$^s$ | $[-6.6, +13.8]$ | $[-11.2, +8.5]$ | $[-6.6, +15.9]$ |
| q8 RTN-W3$^s$ | $[-4.7, +3.9]$ | $[+5.9, +25.9]$ | $[+9.4, +28.2]$ |

**B. Paired differences between upgrade modes**

| Cell | $G - H$ | $F - G$ | $F - H$ |
|---|---|---|---|
| q4 GPTQ-W4 | $[-11.2, +3.6]$ | $[-4.7, +6.1]$ | $[-11.1, +4.4]$ |
| q4 AWQ-style W4 | $[-7.7, +5.1]$ | $[-6.2, +5.4]$ | $[-8.7, +4.7]$ |
| q8 GPTQ-W4 | $[-3.2, +9.3]$ | $[-3.0, +7.1]$ | $[-2.0, +12.5]$ |
| q8 AWQ-style W4 | $[-1.6, +8.6]$ | $[-7.6, +4.2]$ | $[-3.8, +7.5]$ |
| W4 pool | $[-3.1, +3.7]$ | $[-2.8, +3.1]$ | $[-3.5, +4.5]$ |
| q4 KV8 | $[-7.1, +9.7]$ | $[-5.7, +5.3]$ | $[-6.7, +9.5]$ |
| q8 KV8 | $[-3.8, +7.2]$ | $[-2.5, +5.0]$ | $[-3.1, +8.7]$ |
| KV8 pool | $[-3.8, +6.6]$ | $[-2.8, +3.9]$ | $[-3.0, +7.3]$ |
| q4 GPTQ-W3$^s$ | $[-15.0, +4.0]$ | $[-5.2, +10.5]$ | $[-12.2, +6.6]$ |
| q4 KV4-rot$^s$ | $[-13.5, +7.0]$ | $[-1.3, +17.1]$ | $[-6.2, +12.2]$ |
| q8 GPTQ-W3$^s$ | $[-7.7, +11.2]$ | $[-9.1, +9.1]$ | $[-6.8, +11.0]$ |
| q8 KV4-rot$^s$ | $[-3.3, +12.2]$ | $[-7.2, +9.9]$ | $[-2.5, +14.6]$ |
| q8 RTN-W3$^s$ | $[-24.7, -7.1]$ | $[+10.2, +27.8]$ | $[-7.1, +13.3]$ |

Table 8: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 7 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 7 treat the 85 tasks as fixed. $^s$stress cell, 3 blocks.

half's blocks, and ties at the top-20% threshold are broken by a hash. In the reverse direction (rank on the second half, estimate on the first), the 17 fragile tasks turn over 39.0% ($t$ [34.2, 43.9]) and carry 42.3% of the held-out turnover, the other tasks 13.3%; the fragile tasks' net gain is $+3.7$ pp ($t$ $[+0.6, +6.9]$; crossed $[-5.0, +12.4]$) and the rest's $+0.5$. Ranking tasks by BF16 success instead (tertiles on one half) also separates turnover: the hardest tertile turns over 6.7–10.0% and the others 25.4–27.3%, with no net gain resolved under the crossed interval.

## C Telecom Holdout: Protocol, Readings and Replay Audit (supports §4)

This appendix supports the holdout statement of §4: the net estimate is negative, the frozen readings classify it as inconclusive because they had no negative category, and both pre-registered secondary checks passed.

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside $\pm 3$ pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);

**A. Sensitivity to the action and continuation**

| Group | states | action changes | continuation changes | gap (pp) |
|---|---|---|---|---|
| $\tau^2$-bench W4 pool (4 cells) | 5576 | 34.9 [33.6, 36.0] | 29.6 [28.5, 30.7] | +5.2 [+4.4, +6.1] |
| $\tau^2$-bench KV8 pool | 1448 | 26.8 [25.0, 28.6] | 17.0 [14.9, 19.1] | +9.8 [+8.0, +11.6] |
| Telecom q8 W4 pool | 957 | 22.1 [18.2, 25.9] | 21.9 [17.8, 26.1] | +0.1 [−2.1, +2.4] |
| Same-precision W4 (Q′ as B) | 784 | 33.5 [29.8, 37.1] | 28.7 [24.8, 32.2] | +4.9 [+2.9, +6.9] |
| BF16, same blocks | 784 | 35.1 [32.6, 37.4] | 27.8 [23.8, 31.6] | +7.3 [+4.9, +9.7] |
| q8 RTN-W3[s] | 255 | 29.8 [28.1, 31.5] | 37.6 [26.0, 49.3] | −7.8 [−18.1, +2.4] |
| BFCL W4 pool[b] | 2296 | 18.4 [16.4, 20.5] | 11.2 [9.8, 12.6] | +7.2 [+5.8, +8.7] |
| ALFWorld q8 W4 pool[b] | 234 | 44.9 [38.3, 52.4] | 41.5 [34.3, 49.4] | +3.4 [−1.8, +8.4] |

**B. Outcome classes and named patterns**

| Group | unch. | act. only | cont. only | one arm | rev. | joint resc. | resc. undone |
|---|---|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool (4 cells) | 62.4 | 7.9 | 2.7 | 23.8 | 3.2 | 4.0 | 4.8 |
| $\tau^2$-bench KV8 pool | 71.5 | 11.6 | 1.7 | 14.1 | 1.2 | 3.0 | 2.3 |
| Telecom q8 W4 pool | 74.8 | 3.2 | 3.1 | 18.1 | 0.7 | 3.2 | 4.2 |
| Same-precision W4 (Q′ as B) | 64.0 | 7.4 | 2.5 | 22.5 | 3.6 | 2.3 | 5.8 |
| BF16, same blocks | 62.0 | 10.2 | 2.9 | 22.4 | 2.6 | 5.0 | 3.8 |
| q8 RTN-W3[s] | 61.6 | 0.8 | 8.6 | 28.6 | 0.4 | 12.6 | 0.8 |
| BFCL W4 pool[b] | 80.8 | 8.1 | 0.8 | 9.8 | 0.5 | 2.4 | 1.4 |
| ALFWorld q8 W4 pool[b] | 50.0 | 8.5 | 5.1 | 33.4 | 3.0 | 9.4 | 5.5 |

Table 9: Which intervention changes the outcome, in % of divergent states (pools: equal weight per cell). "Action changes": the outcome changes with the action under at least one continuation; "continuation changes": with the continuation controller under at least one action; gap = action only − continuation only. Fixed-pool 95% intervals: block bootstrap for the 20- and 22-block $\tau^2$-bench W4 and KV8 pools; $t$ over per-block ratios for the 6-block telecom and same-precision rows and the 3-block [s]stress cell; [b]one run per item, item bootstrap. Same-precision rows: Q′, a second GPTQ checkpoint of the same bit width, in the B role, and BF16 on the same blocks. New-task (crossed) intervals for the $\tau^2$-bench W4 pool: above.

- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical.

**Replay acceptance on the holdout (post hoc audit).** $Y_{\mathrm{QQ}}$ is the one arm taken from the original episode rather than re-executed, and an exploratory count showed only-QQ successes more often than random labelling would give in two q8 holdout cells. We therefore audited the holdout's acceptance records after the analysis, with a script that outputs only consistency checks and counts. The fixed hash rule selected 135 of the 1202 divergent states for the QQ no-op; all 135 were run and passed, none was missing, failed or non-computable, and no other state was re-run. Beyond the frozen checks (suffix, prompts, termination, outcome, budget counters), the 135 also match on reward, database state, the official reward, the final environment hash against a strict replay of the original episode, and the history prefix. The original QQ and the re-run arms are scored by the same official reward rule: recomputing it from the stored reward information reproduces the stored value for every arm. The audit read the same input files as the frozen analysis, so no repair under the protocol's missing-data rules was needed, and the preregistered result stands as reported. The audit does not explain the only-QQ pattern; a real upgrade that damages the original policy would also produce it. Re-including the two structurally excluded task–cell pairs in all twelve blocks (post hoc sensitivity, frozen estimator) with every unit set to $-1$ or $+1$ gives point estimates of $G_{\mathrm{all}}$ from $-6.68$ [$-10.35$, $-3.01$] to $-2.27$ [$-5.96$, $+1.43$] pp (crossed 90%). This is a range of point estimates: the interval of the most optimistic scenario includes zero. With their observed units kept in the eleven blocks where they exist and only the two missing units set to $-1$ or $+1$, the estimates are $-4.72$ to $-4.35$ pp, each with a crossed 90% interval below zero. The frozen reading is inconclusive in all 36 combinations. Table 12 gives the confirmatory result. Table 13 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is $\leq 0$, and the

**A. Joint rescue**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 3.96 | 3.48 | $+0.47$ [$+0.06$, $+0.87$] | [$-0.33$, $+1.37$] |
| q4 W4 cells | 2541 | 3.22 | 3.56 | $-0.34$ [$-0.83$, $+0.13$] | [$-1.46$, $+0.88$] |
| q8 W4 cells | 3035 | 4.69 | 3.40 | $+1.29$ [$+0.79$, $+1.78$] | [$-0.07$, $+2.74$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.99 | 2.17 | $+0.81$ [$+0.10$, $+1.52$] | [$-0.51$, $+2.19$] |
| Telecom q8 W4 pool | 957 | 3.24 | 3.94 | $-0.70$ [$-1.46$, $+0.05$] | [$-2.47$, $+1.15$] |
| Same-precision W4 (Q′ as B) | 784 | 2.27 | 3.50 | $-1.23$ [$-2.47$, $-0.04$] | [$-2.88$, $+0.51$] |
| BF16, same blocks | 784 | 4.96 | 3.53 | $+1.43$ [$-0.40$, $+3.42$] | [$-0.90$, $+4.26$] |
| Mistral, own BF16 states$^m$ | 464 | 3.4 | 2.9 | $+0.5$ [$-2.4$, $+3.5$] | [$-2.0$, $+3.6$] |

**B. Rescue undone**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 4.80 | 5.78 | $-0.99$ [$-1.58$, $-0.42$] | [$-2.19$, $+0.27$] |
| q4 W4 cells | 2541 | 4.77 | 6.01 | $-1.24$ [$-2.09$, $-0.45$] | [$-2.97$, $+0.69$] |
| q8 W4 cells | 3035 | 4.82 | 5.56 | $-0.74$ [$-1.35$, $-0.07$] | [$-2.34$, $+0.98$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.31 | 4.58 | $-2.28$ [$-2.98$, $-1.58$] | [$-3.62$, $-0.98$] |
| Telecom q8 W4 pool | 957 | 4.18 | 5.13 | $-0.95$ [$-1.70$, $-0.20$] | [$-2.63$, $+0.61$] |
| Same-precision W4 (Q′ as B) | 784 | 5.77 | 5.75 | $+0.02$ [$-2.33$, $+2.25$] | [$-2.66$, $+2.92$] |
| BF16, same blocks | 784 | 3.82 | 6.14 | $-2.32$ [$-4.46$, $-0.19$] | [$-4.68$, $+0.17$] |
| Mistral, own BF16 states$^m$ | 464 | 4.0 | 4.7 | $-0.7$ [$-2.0$, $+0.7$] | [$-3.0$, $+1.9$] |

Table 10: Joint rescue and rescue undone against the conditional label-permutation reference (keeps the number $k$ of successful branches per state), % of divergent states; exploratory, post hoc. 95% intervals: fixed pool (block bootstrap for the 20- and 22-block $\tau^2$-bench pools, $t$ over per-block ratios for fewer than 10 blocks: 6, or 5 for Mistral), new tasks (crossed task×block). Same-precision rows each fork at their own first divergence. $^m$States whose actions differ, on BF16 trajectories (BB is the BF16 original), five blocks; not comparable with Bank P rows.

| Tasks | | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| fragile | est. | 21.1 | 17.6 | 38.7 | $+3.4$ |
| (top 20%) | $t$ | [18.6, 23.5] | [14.3, 21.0] | [34.9, 42.6] | [$-1.0$, $+7.9$] |
| | crossed | [14.8, 28.1] | [12.4, 23.3] | [32.3, 45.7] | [$-7.1$, $+13.7$] |
| rest | est. | 6.5 | 7.2 | 13.7 | $-0.7$ |
| | $t$ | [5.2, 7.8] | [6.2, 8.3] | [12.5, 14.9] | [$-2.7$, $+1.2$] |
| | crossed | [4.4, 9.0] | [4.8, 9.9] | [10.2, 17.5] | [$-4.1$, $+2.4$] |

Table 11: Repeatable task fragility (exploratory, post hoc; four $\tau^2$-bench W4 cells, equal weight). Tasks are ranked by turnover on one hash-defined half of the 20 nonce blocks and measured on the other half (pp of units; 95% intervals, $t$ over held-out blocks and crossed task×block; direction A to B shown). A replication on other nonces of the same tasks, not a prediction for new tasks. Reverse direction: text.

Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues ($-9.09$ pp) and the fewest-faults tercile ($-6.37$ pp).

### C.1 Telecom development tasks

On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\text{all}} = +2.7$ [$-4.7$, $+10.1$] (GPTQ-W4) and $-0.6$ [$-4.9$, $+3.7$] pp (AWQ-style W4), against $-5.6$ and $-6.6$ on the holdout; q4 has no telecom development forks.

## D Reference Experiments in Full: Q′, Common States and Key Order (supports §5)

This appendix supports §5: two-sided flips and continuation dependence also appear without more bits; at common states B's extra action value over Q′ is not resolved and equivalence is not shown; and on the eligible states one key-order change that keeps the tool operation identical also changes outcomes (exploratory).

| Cell (weight) | units | excl. | $G$ | $r$ | $d$ | $\rho_c$ |
|---|---|---|---|---|---|---|
| q4 GPTQ-W4 (1/2) | 396 | 1 | $-3.03$ [$-7.86$, $+1.80$] | $+5.05$ | $+8.08$ | 0.668 |
| q8 GPTQ-W4 (1/4) | 408 | 0 | $-5.64$ [$-9.33$, $-1.94$] | $+4.41$ | $+10.05$ | 0.374 |
| q8 AWQ-style W4 (1/4) | 396 | 1 | $-6.57$ [$-11.98$, $-1.15$] | $+6.31$ | $+12.88$ | 0.644 |
| Weighted ($G_{\text{all}}$) | | | $-4.57$ [$-7.17$, $-1.96$] | $+5.21$ | $+9.77$ | 0.588 |

Table 12: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional ($G_{\text{all}}$, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Alongside interval for the weighted row, $t$ over block means (not decisional): $-4.57$ [$-6.08$, $-3.05$].

| Family | Level | $n$ | $G$ [crossed 90%] | $p$ | Holm $p$ |
|---|---|---|---|---|---|
| step budget[a] | T1 | 138/129/110 | $-0.32$ [$-1.57$, $+0.94$] | .6771 | .6771 |
| | T2 | 124/126/162 | $-2.30$ [$-3.88$, $-0.72$] | .0165 | .0495 |
| | T3 | 123/144/124 | $-1.94$ [$-3.35$, $-0.54$] | .0229 | .0495 |
| persona | Hard | 11 | $-5.99$ [$-12.23$, $+0.26$] | .1147 | .3355 |
| | None | 12 | $-3.65$ [$-7.42$, $+0.13$] | .1118 | .3355 |
| | Easy | 11 | $-4.36$ [$-9.19$, $+0.48$] | .1385 | .3355 |
| BF16 difficulty | 0 | 20 | $-2.35$ [$-4.09$, $-0.62$] | .0257 | .0772 |
| | low | 11 | $-4.54$ [$-8.90$, $-0.19$] | .0860 | .1720 |
| | high | 3 | $-18.75$ [$-38.00$, $+0.50$] | .1091 | .1720 |
| task length | T1 | 9 | $-5.09$ [$-9.81$, $-0.38$] | .0755 | .1511 |
| | T2 | 8 | $-0.78$ [$-4.46$, $+2.90$] | .7270 | .7270 |
| | T3 | 17 | $-6.17$ [$-10.49$, $-1.84$] | .0189 | .0567 |
| issue type | mms | 14 | $-1.15$ [$-3.33$, $+1.04$] | .3880 | .4434 |
| | mobile data | 11 | $-9.09$ [$-14.98$, $-3.20$] | .0111 | .0333 |
| | service | 9 | $-4.40$ [$-10.32$, $+1.52$] | .2217 | .4434 |
| faults | T1 | 19 | $-6.37$ [$-9.91$, $-2.83$] | .0030 | .0091 |
| | T2 | 10 | $-1.94$ [$-7.58$, $+3.69$] | .5704 | .5704 |
| | T3 | 5 | $-2.92$ [$-5.18$, $-0.66$] | .0337 | .0674 |

Table 13: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, $n$ is the number of tasks and $G$ is $G_{\text{all}}$ over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: $n$ is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-style W4), and $G$ is the level's contribution to $G_{\text{all}}$, so the three levels sum to $G_{\text{all}}$. Tertile cut points were fixed on telecom development tasks.

### D.1 Second same-bit-width checkpoint

Per model (final analysis set, $n_{\text{blk}} = 6$), the turnover difference is $\Delta M = +0.6$ [$-3.1$, $+4.3$] pp in q4 and $-2.5$ [$-5.6$, $+0.5$] in q8 (fixed pool, 95%), with crossed intervals [$-5.5$, $+6.9$] and [$-7.3$, $+2.7$]. Within $\pm 3$ pp, equivalence is shown for no per-model $\Delta M$ or $\Delta G_{\text{all}}$; the q8 off-diagonal difference ($+0.40$, 90% [$-2.12$, $+2.92$]) is the only per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate. Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 242/242 (same precision), and no directory was quarantined. At the unit level, Cohen's $\kappa$ between the action-side flip indicators of the two substitutions is 0.43 in q4 and 0.61 in q8; at the task level, cross-design agreement at matched noise is 0.765 of within-design replication.

W3 stress check (§5; $n_{\text{blk}} = 3$): every state passed Q replay, QQ no-op samples passed 31/31 and 39/39, and no directory was quarantined. All branches passed the prefix and environment checks except one of 774 in q8 same precision: a Q′Q′ branch that ended in an agent-side context overflow, for which the harness records no prefix comparison; its environment hash and first replayed prompt

| Reference (status) | States (denominator) | Action replacing Q's (continuation) | Turn-over | Rescues / damages | Net | Paired comparison [interval; 95% unless marked]; reading |
|---|---|---|---|---|---|---|
| Q′: second W4 checkpoint (frozen before outcomes) | each design's own first divergent states; q4, q8 GPTQ-W4; 85 tasks × 6 blocks, 1020 units (% of units) | B (Q) | 16.8 | 8.0 / 8.7 | −0.7 | turnover, B minus Q′ (primary): −1.0; fixed pool [−3.9, +1.9], new tasks [−4.5, +2.7]; not resolved, equivalence not shown; net: −1.6 [−6.2, +3.0], [−6.5, +3.6]; the state sets differ |
| | | Q′ (Q) | 17.7 | 9.3 / 8.4 | +0.9 | |
| Common states (pre-registered at three blocks; extended by amendment) | 510 ordinary q8 GPTQ-W4 states, 85 tasks × 6 blocks, shared by all arms (% of states) | B (Q) | 7.1 | 4.1 / 2.9 | +1.2 | B's action beyond Q′'s, Q continues: +1.0; two-look 97% $t$ [−4.2, +6.2], bootstrap [−1.4, +3.5]; unadjusted $t$ [−3.5, +5.4], bootstrap [−1.2, +3.1]; not resolved |
| | | Q′ (Q) | 8.0 | 4.1 / 3.9 | +0.2 | |
| | | B (B; vs. QB) | 9.0 | 5.3 / 3.7 | +1.6 | B's action beyond Q′'s, B continues: +0.8; two-look 97% [−2.2, +3.8], [−2.0, +3.7]; unadjusted $t$ and bootstrap [−1.8, +3.3]; not resolved; equivalence: unadjusted 90% $t$ [−1.2, +2.8] inside ±3, two-look 94% $t$ [−1.6, +3.2] not; no equivalence claimed |
| | | Q′ (B; vs. QB) | 8.6 | 4.7 / 3.9 | +0.8 | |
| Key order (exploratory; code frozen before outcomes) | 300 q8 GPTQ-W4 first divergent states whose action is one tool call with at least two argument keys, 41 tasks (% of states) | B (Q) | 9.0 | 3.3 / 5.7 | −2.3 | turnover, reversed keys minus B: −2.3 [−8.4, +2.9]; not resolved; no equivalence margin |
| | | Q, keys reversed (Q) | 6.7 | 3.3 / 3.3 | 0.0 | |

Table 14: Reference experiments with all registered intervals (full version of Table 1). Each row group has its own state source and denominator; comparisons are paired only within a group, and the references are never subtracted from B's turnover. Turnover, rescues, damages and net compare the replacing action with Q's action under the same continuation: against QQ when Q continues, and against Q's action followed by B (QB) when B continues. Intervals are those each protocol registers: Q′, fixed pool ($t$ over 6 block means) and new tasks (crossed); common states, $t$ over block means and task-cluster bootstrap, a reading needs both, under the two-look (Pocock) reading that the protocol designates confirmatory and, unadjusted, at the frozen levels (Appendix D.2); key order, the wider of block and crossed bootstrap.

match the frozen state, and by the protocol it counts as a failure, not an exclusion.

**Equivalence tests (W4).** The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI $[-3.26, +1.30]$). No precision-specific excess was detected, but equivalence within ±3 pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\text{all}} = -1.57$ $[-5.17, +2.04]$, $p = 0.230$) and, descriptively, for the off-diagonal share ($-1.210$ $[-4.09, +1.67]$, $p = 0.133$, as the mean of per-block ratios, which the interval and test use; $-1.216$ as the ratio of block sums).

**Units affected by both, or by only one, substitution.** At W4, 107 units flip under both substitutions, 64 only under B and 74 only under the second checkpoint. A within-task block shuffle, which keeps each task's flip propensity, gives 76 [69, 84] units flipping under both, so the two substitutions share affected units beyond what flip-prone tasks explain. The overlap lies in action replacement: over the 1020 W4 units, Cohen's $\kappa$ between the indicators "B's action rescues" and "the second checkpoint's action rescues" is $\kappa_r = 0.60$, and for damages $\kappa_d = 0.54$. When B also takes over the continuation, the affected units are largely substitution-specific: rescue sets under the continuation-only contrast agree with $\kappa = 0.20$, and 70% of continuation-dependent units are continuation-dependent under one substitution only. The difference in action-side net

value ($\Delta G_{\text{all}}$) comes only from the substitution-specific units and is not resolved ($-1.6$ pp; fixed pool [$-6.2$, $+3.0$], new tasks [$-6.5$, $+3.6$]).

**Full substitution (W4).** Under full substitution B flips more units than Q′ (193 vs. 149; $+4.3$ [$+0.4$, $+8.3$] pp, one of several contrasts); the action-side net difference ($\Delta G_{\text{all}}$, B's action against Q′'s under Q's continuation, not an $F$ difference) comes only from substitution-specific units and is not resolved ($-1.6$ [$-6.2$, $+3.0$] pp). Only value measures on the same units reveal this structure.

**W3 stress pair.** The protocol's complete W3 panel (Q GPTQ-W3, Q′ a second W3 calibration draw; q4 and q8, $n_{\text{blk}} = 3$, blocks q3–q5, 510 units) is reported next to W4, never pooled with it. The same-precision substitution produces 17.3 pp of turnover, the BF16 substitution 18.8 pp; off-diagonal shares are 27.6% and 27.3%. $\Delta M = +1.6$ pp (95% [$-0.66$, $+3.80$]) is unresolved: the preregistered $\pm 3$ pp equivalence test fails ($p = 0.055$); that the 90% interval lies just above zero does not change this. $\Delta G_{\text{all}}$ is equivalent within $\pm 3$ pp only in the preregistered pooled population ($+1.2$, $p = 0.017$); per model it is $+3.1$ (q4) and $-0.8$ pp (q8). The off-diagonal difference is $-0.288$ (ratio of block sums) or $-0.307$ (mean of per-block ratios, used by the interval and test: 90% [$-11.0$, $+10.4$], $p = 0.27$). On the action side ($G$), rescued and harmed task×block units overlap ($\kappa = 0.60$, $0.68$); the designs may first diverge at different decisions, so this is unit overlap, not a same-state replication. Of 122 units whose $G$ outcome flips, 62 flip under both substitutions, 34 only under BF16 and 26 only under the other W3 checkpoint: 49% respond to only one. Flip volume, affected units and net upgrade value must be measured separately.

### D.2 Common states: extra value of the BF16 action

**Common-state comparison: details.** Above, each design selects its own divergent states. A comparison pre-registered at three blocks removes that difference: on 510 ordinary q8 states (85 tasks × 6 blocks, sampled by a fixed rule), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\text{B}, c) - V(\text{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both intervals exclude zero (Holm), equivalence only if both lie within $\pm 3$ pp (two one-sided tests). The design first ran three blocks, and that first look was unresolved (under Q's continuation $K = -0.4$ pp, 95%: $t$ [$-9.3$, $+8.5$], bootstrap [$-3.5$, $+2.7$]; under B's $+2.4$ pp, [$-3.5$, $+8.2$], [$-0.8$, $+5.5$]). Three more blocks were then added. That decision followed the unresolved first look: a dated amendment fixed the extension and the two-look rule after that look and before any extension run had produced output, and the protocol designates this two-look reading (Pocock levels from a normal approximation: about 97% intervals for a difference and 94% for equivalence) as the confirmatory one. The Pocock constant is not calibrated for small-$k$ $t$ intervals, the task bootstrap or the joint rule, so the two-look reading carries no exact error guarantee. Because the extension was chosen after the first look, even the two-look reading cannot fully protect the error rate conditional on the observed first look (protocol D2.2). We therefore print it next to the unadjusted reading at the frozen levels and claim neither an advantage nor equivalence unless both agree. After the fixed extension (six blocks), B's action is not shown to be worth more than Q′'s, and equivalence is not established. Under Q's continuation, $K = +1.0$ pp (95%: $t$ [$-3.5$, $+5.4$], bootstrap [$-1.2$, $+3.1$]; two-look 97%: [$-4.2$, $+6.2$], [$-1.4$, $+3.5$]). Under B's, $K = +0.8$ pp (95%: [$-1.8$, $+3.3$] for both; two-look 97%: [$-2.2$, $+3.8$], [$-2.0$, $+3.7$]). For equivalence under B's continuation, the ordinary 90% intervals lie within $\pm 3$ pp ($t$ [$-1.2$, $+2.8$], bootstrap [$-1.4$, $+2.9$]) and the 94% intervals of the protocol-designated two-look confirmatory reading do not ($t$ [$-1.6$, $+3.2$]); because the extension followed the first look, we do not claim equivalence. Without the 26 states at which earlier studies had already produced outcomes, $K$ is $+0.5$ and $+0.8$ pp, also not resolved, and the unadjusted reading no longer shows equivalence under B's continuation either; BF16's own outcome at the extension states was also known in aggregate before the amendment (protocol D2.1). Under Q's continuation, B's action flips $M = 7.1\%$ of outcomes (rescue 4.1, damage 2.9) for a net $\Delta = +1.2$ pp (95%: $t$ [$-1.3$, $+3.6$]); Q′'s flips 8.0% (4.1, 3.9) for $+0.2$ pp ([$-2.3$, $+2.7$]; within $\pm 3$ pp by two one-sided tests). A second W4 checkpoint's action departs from Q's at least as often as B's does (agreement with Q's action 53.9% against 55.7%; B with Q′ 57.3%). The interaction of $K$ with the continuation is $-0.2$ pp (95%: $t$ [$-4.5$, $+4.1$], bootstrap

[−2.7, +2.4]; secondary, not resolved). *q4, a separate prospective analysis.* The same design on Qwen3-4B planned three blocks. The frozen quarantine rule removed f01 (below), leaving 170 states. $K = -1.2$ pp under Q's continuation (bootstrap 95% [−5.9, +3.5]) and −2.4 pp under B's ([−6.5, +1.2]), both not resolved. With two blocks the $t$ interval has one degree of freedom. Under Q's continuation the two block means happen to coincide, so it collapses to a point. That is a coincidence of two values, not a precise measurement or zero variance, and the reading rests on the bootstrap. *Identity checks.* Across both models, 1,238 of 1,240 reference and implicit identity checks passed. Both failures are one q4 state. There the substituted action was semantically identical to the original after parsing, but its token sequence differed in the order of keys inside a tool-call argument. The canonical comparison treats such actions as equal, but the chat template renders the order, so the next prompt differed and the continuation could not be a replay. Every check with a token-identical action passed (869/869). As the frozen rule requires, the block was removed. Changing outcomes is thus not improving them; this does not make precision irrelevant or the two actions equivalent.

**Common states, BF16 against Q′, by component.** At the same states and continuation, BF16's extra rescues over Q′ on q8 are +0.3 pp and its extra damages −0.6 pp, with the net $K = +0.9$ unresolved; each component lies within ±3 pp under a 90% two one-sided test (exploratory, post hoc; the margin is wide relative to rescue rates of 3–5%). q4 is unresolved.

### D.3 Key-order control

**Design.** Protocol `KEYORDER_PERTURB_PROTOCOL.md` (ko-v1, exploratory control). Eligible states are first tool-structure divergences of q8 GPTQ-W4 (Bank P) where Q's action is a tool call with at least two top-level argument keys, reversing them leaves the parsed call and the environment effects unchanged, and the next prompt tokens differ (300 of 1473 divergent states, 41 tasks). The arm KQ forces the reversed action and lets Q continue; QQ and BQ are reused from Bank P. Eligibility was fixed without outcomes; the analysis code (sha256-recorded) was frozen before any KQ outcome was read. All four tasks passed acceptance (KQ checks 300/300, shadow 300/300, QQ no-op 35/35, no errors or context overflows). Two tasks were restarted with resume during the run (Amendment 2026-10-05g).

**Results.** Conditional on eligible states (pooled ratio; intervals are the wider of block and crossed bootstrap), KQ changes 6.7% [0.8, 14.4] of outcomes and B 9.0% [2.9, 18.2]; KQ has 10 rescues and 10 damages, B 10 and 17, and the flips coincide on 5 rescues and 7 damages. Over the full 1700-unit pool, KQ contributes 1.2 pp [0.1, 2.8] of turnover against 17.9 pp for the BF16 swap on all divergent states.

## E Trigger, Oracle and KL Detail (supports §6)

This appendix supports §6: the frozen Q-only trigger enriches outcome-changing decisions but does not select net-beneficial upgrades, and token KL tracks how often actions differ, not net value (exploratory, post hoc).

### E.1 The frozen trigger and its headroom

Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell × state rows, 58 flips), teacher-forcing both models on the generated ids of $a_{\mathrm{B}}$ and $a_{\mathrm{Q}}$. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

**Uncertainty is associated with divergence.** Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified). Among already divergent U1 states, no Q-visible statistic has an interval excluding AUC 0.5; small-subset associations vary across state sources. This development diagnosis motivated the frozen margin rule, whose validation reuses the tasks with fresh nonce blocks. On the secondary validation population, ordinary states from BF16 trajectories (U2B; f02/f03, 1020 rows, 75 outcome-changing states), the frozen rule captures 4, 11 and 35 outcome-changing states against 3.81, 8.35 and 16.07 expected at random ($p = .5404$ and .1995 at 5 and 10%; the 20% excess is resolved), with net gains of −2, −1 and +5 against random +0.07, +1.10 and +2.48.

**A. Frozen trigger and matched hindsight oracle**

| Budget | sel. | oracle r/d/net (pp) | frozen r/d/net (pp) | net ratio |
|---|---|---|---|---|
| 5% | 50 | 34/0/+34 (+3.33) | 6/2/+4 (+0.39) | 4/34 |
| 10% | 114 | 38/0/+38 (+3.73) | 7/5/+2 (+0.20) | 2/38 |
| 20% | 222 | 38/0/+38 (+3.73) | 16/11/+5 (+0.49) | 5/38 |

**B. Post hoc KL reference (requires BF16)**

| Budget | sel. | net vs. random ($p$; Holm) | r/d | turnover vs. random ($p$) |
|---|---|---|---|---|
| 5% | 72 | −1 vs. +0.45 (.8337; 1.0000) | 5/6 | 11 vs. 4.58 (.0038) |
| 10% | 126 | +1 vs. +0.82 (.5483; 1.0000) | 8/7 | 15 vs. 7.54 (.0048) |
| 20% | 257 | +3 vs. +1.59 (.3984; 1.0000) | 18/15 | 33 vs. 16.73 ($< 10^{-4}$) |

Table 15: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Net ratio: frozen net over oracle net (not a share of rescues; the frozen rule selects 6, 7 and 16 of the oracle's rescues). Random: matched-count per-cell random-trigger expectation; $p$: exact one-sided hypergeometric, Holm over the three budgets.

### E.2 Matched references for the frozen trigger

Table 15 compares the frozen trigger of §6 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per $a_{\mathrm{Q}}$ token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and $a_{\mathrm{Q}}$ at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations over the $a_{\mathrm{Q}}$ tokens were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; the maximum was chosen, and all of f02/f03 is reported. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

### E.3 Fidelity, sensitivity and signed value

**KL tracks action divergence and turnover, not net value.** On ordinary BF16-sourced states shared by every method (U2B), we compare methods by $D_{cc} = Y_{\mathrm{QB}} - Y_{\mathrm{BB}}$, Q's action against B's under the shared BF16 continuation, with turnover $M_{cc}$ (Appendix F.5). These comparisons evaluate complete quantization configurations, including their calibration pipelines, rather than isolated algorithmic components. Mean teacher-forced approximate token $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ tracks how often actions differ, not the direction of their consequences (correlations below), consistent with KL measuring the size of a disagreement rather than its direction (Nikolić et al., 2026). SpinQuant-W4A4 has lower KL than QuaRot-W4A4 (0.165 against 0.199), yet their pre-registered task-value contrast is unresolved (+1.2 pp, crossed [−4.7, +7.1]; Table 16).

The following decomposition was added after panel results were read and is exploratory. On each cell's admitted states, exact action equality and deterministic continuation give

$$M_{cc} = P(a_{\mathrm{Q}} \neq a_{\mathrm{B}})\, P(Y_{\mathrm{QB}} \neq Y_{\mathrm{BB}} \mid a_{\mathrm{Q}} \neq a_{\mathrm{B}}).$$

B's rescues $r_{cc} = P(Y_{\mathrm{BB}} = 1, Y_{\mathrm{QB}} = 0)$ and damages $d_{cc} = P(Y_{\mathrm{BB}} = 0, Y_{\mathrm{QB}} = 1)$ give $D_{cc} = d_{cc} - r_{cc}$. The decomposition separates how often an action changes from what a changed action does; it is diagnostic rather than a mechanism.

**Approximate KL.** Both models are teacher-forced on identical prompt ids and B's action tokens. At each token, $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ uses the union of both top-20 sets and the forced token, plus one tail bucket per model. An in-union token missing from a model's top-20 receives min(the 20th probability, tail mass divided by the number of missing tokens). The state score averages over B's action tokens. This coarse distribution is not the full token distribution.

| SpinQuant − QuaRot | difference [fixed pool] | [new tasks] |
|---|---|---|
| KL ($\times 10^{-3}$) | $-34$ [$-57$, $-11$] | [$-56$, $-11$] |
| Action difference (pp) | $+0.8$ [$-11.4$, $+13.0$] | [$-8.2$, $+9.4$] |
| Flip given difference (pp) | $+2.7$ [$-10.2$, $+15.3$] | [$-7.5$, $+11.4$] |
| $D_{cc}$ (pp) | $+1.2$ [$-3.9$, $+6.2$] | [$-4.7$, $+7.1$] |

Table 16: Paired diagnostic on 255 identical W4A4 states; 95% intervals, $t$ over three block differences and crossed task×block bootstrap. The KL reduction is resolved; the differences in the task quantities remain unresolved. The $D_{cc}$ row is primary contrast P2.

**Across and within configurations.** Across eleven q8 cells, KL's descriptive Spearman correlation is 0.99 with the action-difference rate, 0.925 with $M_{cc}$, 0.81 with rescues and 0.93 with damages. It is 0.46 with the flip rate conditional on a difference and $-0.44$ with $D_{cc}$. Removing the three unquantized path controls (eight configurations) gives 0.98, 0.64 and $-0.52$ for the action-difference rate, conditional flip rate and $D_{cc}$. These configurations are not exchangeable draws. Within cells, the mean AUC is 0.93 for an action difference, 0.46 for a flip conditional on a difference and 0.55 for a rescue conditional on a flip; the last uses 3–39 flips per cell. These diagnostics establish no reliable predictor of net upgrade value.

# F Scope Checks: Ordinary States, Other Environments, a Second Family and Other Configurations (supports Table 2)

This appendix supports Table 2: it reports each scope check in full and states where a check does not support extending the main-panel statements.

## F.1 Ordinary states and the user simulator

**Ordinary states turn over less, mostly through text under our simulator.** At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; $t$ intervals over block means), $M$ per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within $\pm 2.8$ pp (U1, U2B) and $\pm 5.0$ pp (U2Q), and every interval covers zero (Appendix F.2). Identical actions never change the outcome, as determinism implies. Reused arms rest on canonical equality. All 4,114 such reuses were rendered with each model's chat template; 4,113 are token-identical. The remaining pair differs only in JSON key order, and its two cross arms, measured separately, gave the reused outcomes. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii), the same tool structure with different text; 26 involve a different tool name, call count or argument (Table 18). The rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more.

**Sensitivity to the user simulator.** Text actions act through the user simulator, and changing the simulator moves $\tau$-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels.

**Sensitivity of the four-branch results.** Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms

| Cell | U1 ($n$=255) | U2B ($n$=255) | U2Q ($n$=60) |
|---|---|---|---|
| q4 GPTQ-W4 | .090 (11/12) | .082 (10/11) | .083 (4/1)* |
| q4 AWQ-style W4 | .082 (7/14) | .078 (7/13) | .083 (3/2) |
| q4 KV8 | .047 (4/8) | .039 (5/5) | .017 (0/1) |
| q8 GPTQ-W4 | .047 (8/4) | .067 (9/8) | .050 (3/0)* |
| q8 AWQ-style W4 | .086 (14/8) | .090 (15/8) | .067 (3/1) |
| q8 KV8 | .035 (5/4) | .031 (6/2) | 0 (0/0) |

Table 17: Bank U, blocks f01–f03: per-state turnover $M$ (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). $G$ lies within $\pm 0.028$ in U1 and U2B and within $\pm 0.050$ in U2Q, and every interval includes 0. Intervals are $t$ intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

| Model:sub | (i) identical | (ii) same tool structure, text | (iii) different tool or args | (iv) act vs. talk |
|---|---|---|---|---|
| q4:U1 | 399: 0 | 309: 45 (.146) | 19: 1 (.053) | 38: 10 (.263) |
| q8:U1 | 483: 0 | 202: 28 (.139) | 47: 11 (.234) | 33: 4 (.121) |
| q4:U2B | 430: 0 | 277: 44 (.159) | 24: 4 (.167) | 34: 3 (.088) |
| q8:U2B | 470: 0 | 199: 25 (.126) | 40: 9 (.225) | 56: 14 (.250) |
| q4:U2Q | 108: 0 | 63: 10 (.159)* | 5: 0 | 4: 1 (.250) |
| q8:U2Q | 129: 0 | 33: 6 (.182)* | 11: 1 (.091) | 7: 0 |
| unique rows | 1943: 0 | 1045: 150 (.144) | 143: 26 (.182) | 169: 32 (.189) |

Table 18: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. The 58 f01 flips scored in Appendix E.1 include the two duplicated U2Q rows.

five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change $-6.0$ $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 7 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings.

### F.2 Bank U per Sub-Bank

Table 17 gives the per-state turnover behind Appendix F.1, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

### F.3 Models compared on their own ordinary states

**Mistral population, scoring and coverage.** The Mistral panel applies the same row rule to Mistral's own BF16 trajectories on the 85 retail and airline tasks. Seven BF16 source rows are missing because the source episode ended in an invalid tool call produced by the model itself (airline task 44 in all six blocks, airline task 13 in f01). A dated amendment, fixed from the BF16 source runs before any Mistral fork output existed, excludes both tasks in every block and cell, leaving 83 tasks. The selection was outcome-blind, but BF16 Mistral fails these tasks structurally, so the estimates describe this conditional population, not the 85-task panel, and are not compared with q8 as if the populations were equal. A branch that ended because the harness could not accept the agent's own message (an empty or invalid tool name, or an empty message) is scored as a failure, as in end-to-end evaluation. This rule was added after the Mistral phase-1 results had been read; without it the Mistral primary comparison is not estimable from phase 1 (one complete block). The analysis without the rule (which leaves such an arm missing and fails its block) and one applying the rule only to data not yet read when it was fixed are reported alongside;

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| GPTQ-W4 | 6 | 498 | $-0.8$ [$-5.2$, $+3.6$] [$-5.8$, $+4.2$] | $+1.4$ [$-3.0$, $+5.8$] [$-4.0$, $+6.8$] | $+0.0$ [$-4.2$, $+4.2$] [$-6.0$, $+6.0$] |
| RTN-W4 | 5 | 415 | $+0.7$ [$-2.7$, $+4.2$] [$-3.6$, $+5.5$] | $+1.9$ [$-3.8$, $+7.7$] [$-4.6$, $+8.0$] | $+2.4$ [$-2.2$, $+7.0$] [$-3.9$, $+8.4$] |

Table 19: Mistral-Small-3.2-24B on its fixed 83-task conditional population. pp; fixed-pool 95% $t$ intervals, then crossed task×block 95% intervals. Missing argument-JSON errors leave RTN f04 excluded, whereas model-produced invalid tool names are scored as failures under the dated amendment above. None of the three mode intervals excludes zero.

both admit the same three blocks for GPTQ − RTN ($+1.2$ [$-13.8$, $+16.2$]). Coverage: GPTQ-W4 has 498 states on six blocks and RTN-W4 415 states on five. Full estimates on the 83 tasks (GPTQ-W4 on six blocks, RTN-W4 on five; Table 19): in GPTQ-W4, B's action rescues $r = 4.6$ [2.4, 6.8] and damages $d = 5.4$ [2.4, 8.5] pp of states ($M = 10.0$ [7.0, 13.1]; fixed-pool 95%), and no mode's net value is resolved ($G = -0.8$ [$-5.2$, $+3.6$], $H = +1.4$ [$-3.0$, $+5.8$], $F = +0.0$ [$-4.2$, $+4.2$]). RTN-W4 is similar ($M = 10.4$ [6.3, 14.4]). The pre-registered contrast GPTQ − RTN is not resolved: $\Delta D_{cc} = +2.2$ [$-3.5$, $+7.8$] pp, $p = .35$, on five blocks, two of which are admitted only by a scoring rule adopted after an interim look (above).

### F.4 Replacement and four-branch estimands on BFCL and ALFWorld (supplementary)

Table 20 gives the replacement quantities of §4 ($r$, $d$, $M$ and $G_{\text{all}}$ over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The four-branch columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{\text{QQ}} = 0$ on every state, so $\Delta_{\text{Q}}$ is almost constant and their off-diagonal mass is $H$. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%). The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ [$-43.5$, $-25.4$] pp; KV4 cells reduce to $H$.

### F.5 Quantization methods on shared ordinary states

These tests ask whether numerical error locates changed actions, consequential changes and beneficial changes. Methods share ordinary BF16-sourced states (U2B); the separate Mistral population is specified in Appendix F.3.

**Contrasts.** Methods are compared on shared states by $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ ($= -\Delta_{\text{B}}$): Q's action against B's under the common BF16 continuation. Because BB is the shared BF16 original, two methods differ on the same states only through $Y_{\text{QB}}$. One pre-registered Holm family holds three contrasts on $D_{cc}$, with $t$ intervals over block-mean differences: P1, QuaRot-W4 − GPTQ-W4; P2, SpinQuant-W4A4 − QuaRot-W4A4; P3, KIVI-2 − KV8. Unless marked crossed, intervals for these panels in §6 and §7 are fixed-pool 95% ($t$ over block means, as fewer than 10 blocks are complete); other new-task (crossed) intervals are in the analysis outputs.

**Numerical quality.** QuaRot-W4 against GPTQ-W4 is $-0.4$ [$-10.7$, $+9.9$] pp (P1). The matched SpinQuant-W4A4 contrast is P2 (complete family: Table 21; paired diagnostic components: Table 16). Its third block rests on a one-time re-run of arms lost to a user-simulator overload; on the two blocks without it the contrast is $+2.4$ pp (strict view).

**Upgrade modes on the q8 panel.** On the q8 ordinary-state panel, B's action alone ($G$) is resolved in no cell. Fixed-pool intervals for B's takeover $H$ or full substitution $F$ exclude zero in weight-only (AWQ-style W4, QuaRot-W4), W4A4 and KV8 cells, and new-task intervals still do in both W4A4 cells and KV8 (unadjusted), so resolved value is not confined to quantized activations. It is largest in SpinQuant-W4A4 ($H = +11.8$ [6.7, 16.8], $F = +13.7$ [9.3, 18.2] pp).

**KIVI.** KIVI (Liu et al., 2024) quantizes keys per channel and values per token. KIVI-2 against KV8 is not resolved ($-0.6$ [$-38.0$, $+36.8$] pp), nor is KIVI-4 against KV8 ($+1.6$ [$-2.9$, $+6.0$] pp). Our implementation is a replay-compatible variant: at every decode step it reconstructs the quantized view

**A. Full item pool**

| Cell | items | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 788 | 6.3 | 4.3 | 10.7 | $+2.0$ $[-0.3, +4.3]$ |
| BFCL q4 AWQ-style W4 | 788 | 5.7 | 9.5 | 15.2 | $-3.8$ $[-6.5, -1.3]$ |
| BFCL q8 GPTQ-W4 | 788 | 2.9 | 2.3 | 5.2 | $+0.6$ $[-0.9, +2.3]$ |
| BFCL q8 AWQ-style W4 | 788 | 4.1 | 2.7 | 6.7 | $+1.4$ $[-0.3, +3.3]$ |
| BFCL W4 pool | | | | | $+0.1$ $[-1.0, +1.2]$ |
| ALFWorld q8 GPTQ-W4 | 134 | 9.0 | 12.7 | 21.6 | $-3.7$ $[-11.2, +4.5]$ |
| ALFWorld q8 AWQ-style W4 | 134 | 10.4 | 14.2 | 24.6 | $-3.7$ $[-11.9, +4.5]$ |
| ALFWorld W4 pool | | | | | $-3.7$ $[-9.7, +1.9]$ |
| BFCL q4 GPTQ-W3$^s$ | 591 | 6.1 | 3.6 | 9.6 | $+2.5$ $[+0.2, +5.1]$ |
| BFCL q8 GPTQ-W3$^s$ | 788 | 5.2 | 4.1 | 9.3 | $+1.1$ $[-1.0, +3.3]$ |
| BFCL q4 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| BFCL q8 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| ALFWorld q8 GPTQ-W3$^s$ | 134 | 6.7 | 3.0 | 9.7 | $+3.7$ $[-1.5, +9.7]$ |
| ALFWorld q8 KV4$^s$ | 134 | 0.0 | 0.0 | 0.0 | $+0.0$ $[+0.0, +0.0]$ |

**B. First divergent states**

| Cell | states | $G$ | $H$ | $F$ | $G - H$ | off-diag. % |
|---|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 573 | $+2.8$ $[-0.4, +5.9]$ | $+1.4$ $[-0.7, +3.5]$ | $+4.2$ $[+0.9, +7.2]$ | $+1.4$ $[-1.8, +4.5]$ | 11.7 |
| BFCL q4 AWQ-style W4 | 642 | $-4.7$ $[-7.8, -1.5]$ | $-1.1$ $[-3.6, +1.4]$ | $-1.4$ $[-4.8, +1.7]$ | $-3.6$ $[-6.7, -0.2]$ | 17.1 |
| BFCL q8 GPTQ-W4 | 533 | $+0.9$ $[-1.3, +3.3]$ | $+1.5$ $[+0.0, +3.2]$ | $+2.1$ $[-0.4, +4.6]$ | $-0.6$ $[-3.1, +2.0]$ | 4.7 |
| BFCL q8 AWQ-style W4 | 548 | $+2.0$ $[-0.4, +4.7]$ | $+1.8$ $[+0.0, +3.7]$ | $+3.6$ $[+0.9, +6.5]$ | $+0.2$ $[-2.6, +3.1]$ | 7.8 |
| BFCL W4 pool | | $+0.3$ $[-1.2, +1.8]$ | $+0.9$ $[-0.2, +2.0]$ | $+2.1$ $[+0.3, +3.9]$ | $-0.6$ $[-2.3, +1.0]$ | 10.3 |
| ALFWorld q8 GPTQ-W4 | 115 | $-4.3$ $[-13.3, +5.0]$ | $+7.0$ $[-2.6, +16.7]$ | $+9.6$ $[-0.9, +19.7]$ | $-11.3$ $[-20.9, -1.8]$ | 40.0 |
| ALFWorld q8 AWQ-style W4 | 119 | $-4.2$ $[-13.6, +4.9]$ | $+5.0$ $[-2.6, +12.2]$ | $+4.2$ $[-5.9, +13.7]$ | $-9.2$ $[-18.8, +0.0]$ | 32.8 |
| ALFWorld W4 pool | | $-4.3$ $[-11.1, +2.2]$ | $+6.0$ $[-0.5, +12.6]$ | $+6.9$ $[-2.1, +15.5]$ | $-10.3$ $[-17.7, -2.9]$ | 36.4 |
| BFCL q4 GPTQ-W3$^s$ | 540 | $+2.8$ $[+0.2, +5.5]$ | $+5.2$ $[+2.9, +7.7]$ | $+8.9$ $[+5.7, +12.2]$ | $-2.4$ $[-5.7, +0.7]$ | 13.0 |
| BFCL q8 GPTQ-W3$^s$ | 696 | $+1.3$ $[-1.1, +3.7]$ | $+1.4$ $[-0.1, +2.9]$ | $+3.4$ $[+0.9, +6.0]$ | $-0.1$ $[-2.6, +2.4]$ | 8.3 |
| BFCL q4 KV4$^s$ | 787 | $+0.3$ $[+0.0, +0.6]$ | $+2.9$ $[+1.9, +4.1]$ | $+20.3$ $[+17.6, +23.0]$ | $-2.7$ $[-3.9, -1.5]$ | 20.5 |
| BFCL q8 KV4$^s$ | 788 | $+0.3$ $[+0.0, +0.6]$ | $+1.8$ $[+1.0, +2.7]$ | $+14.5$ $[+12.1, +17.0]$ | $-1.5$ $[-2.5, -0.6]$ | 14.7 |
| ALFWorld q8 GPTQ-W3$^s$ | 130 | $+3.8$ $[-1.5, +9.8]$ | $+38.5$ $[+29.8, +46.9]$ | $+42.3$ $[+33.6, +50.8]$ | $-34.6$ $[-43.5, -25.4]$ | 30.8 |
| ALFWorld q8 KV4$^s$ | 134 | $+0.0$ $[+0.0, +0.0]$ | $+38.1$ $[+30.6, +45.5]$ | $+53.0$ $[+44.8, +61.2]$ | $-38.1$ $[-45.5, -30.6]$ | 26.9 |

Table 20: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue $r$, damage $d$, turnover $M$ (% of all items) and net $G_{\text{all}}$ (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$. Pools: equal weight per cell. $^s$Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to $H$.

that official KIVI's token-by-token decode would hold at that position from the full-precision cache, rather than persisting a compressed cache across turns, so every token, prompt or generated, attends with that view. The KIVI-2 contrast (P3) rests on two blocks: the third lost one arm to a server time-out that recurred on a single-request re-run, and both admitted blocks depend on the re-run of arms lost to server failures (strict rule: not estimable). Net of the kernel path, the quantization effect is $+0.6$ $[-51.7, +52.9]$ (KIVI-2) and $+2.4$ $[-5.4, +10.1]$ pp (KIVI-4), and the unquantized kernel path itself has $D_{cc} = -1.2$ $[-6.2, +3.9]$.

**All admitted ordinary-state upgrade effects.** Table 22 reports all three upgrade modes for ev-

| Contrast | blocks | $\Delta D_{cc}$ [fixed pool] | [new tasks] | Holm $p$ |
|---|---|---|---|---|
| P1: QuaRot − GPTQ W4 | 3 | $-0.4$ [$-10.7$, $+9.9$] | [$-5.9$, $+5.5$] | 1.0 |
| P2: SpinQuant − QuaRot W4A4 | 3 | $+1.2$ [$-3.9$, $+6.2$] | [$-4.7$, $+7.1$] | 1.0 |
| P3: KIVI-2 − KV8 | 2 | $-0.6$ [$-38.0$, $+36.8$] | [$-7.1$, $+7.1$] | 1.0 |

Table 21: Complete primary method-comparison family on shared states, pp; fixed-pool and new-task 95% intervals. All three comparisons remain unresolved. Admission and re-run dependencies are described in the text.

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| AWQ-style W4 | 3 | 255 | $+2.7$ [$-4.6$, $+10.1$] [$-2.4$, $+9.0$] | $+4.3$ [$-0.1$, $+8.8$] [$-1.2$, $+11.0$] | $+5.5$ [$+1.0$, $+10.0$] [$-0.8$, $+12.5$] |
| GPTQ-W4 | 3 | 255 | $+0.4$ [$-7.0$, $+7.7$] [$-4.3$, $+5.9$] | $+0.8$ [$-5.3$, $+6.9$] [$-5.9$, $+7.5$] | $+3.5$ [$-2.3$, $+9.4$] [$-3.9$, $+10.6$] |
| KIVI-16 path | 3 | 255 | $+0.0$ [$-2.9$, $+2.9$] [$-2.7$, $+2.7$] | $+1.2$ [$-6.6$, $+8.9$] [$-3.5$, $+5.9$] | $+2.4$ [$-0.6$, $+5.3$] [$-2.4$, $+7.1$] |
| KIVI-2 | 2 | 170 | $-2.9$ [$-25.4$, $+19.5$] [$-8.2$, $+2.4$] | $+2.9$ [$-4.5$, $+10.4$] [$-3.5$, $+10.0$] | $+2.9$ [$-34.4$, $+40.3$] [$-5.9$, $+10.6$] |
| KIVI-4 | 3 | 255 | $+0.0$ [$-5.1$, $+5.1$] [$-3.5$, $+4.3$] | $+2.0$ [$-1.4$, $+5.3$] [$-3.5$, $+7.5$] | $+0.8$ [$-8.1$, $+9.7$] [$-5.9$, $+7.1$] |
| KV8 | 3 | 255 | $+1.6$ [$-0.1$, $+3.3$] [$-1.2$, $+4.7$] | $+5.1$ [$+0.6$, $+9.6$] [$+0.4$, $+9.8$] | $+5.5$ [$+1.0$, $+10.0$] [$+0.0$, $+10.6$] |
| QuaRot-W16 path | 3 | 255 | $+0.0$ [$-5.1$, $+5.1$] [$-3.5$, $+3.1$] | $+1.6$ [$-14.5$, $+17.7$] [$-5.9$, $+9.4$] | $+2.4$ [$-8.2$, $+12.9$] [$-4.3$, $+9.0$] |
| QuaRot-W4 | 3 | 255 | $+0.8$ [$-5.3$, $+6.9$] [$-3.5$, $+5.9$] | $+1.6$ [$-1.8$, $+4.9$] [$-4.3$, $+7.5$] | $+4.7$ [$+1.8$, $+7.6$] [$-2.0$, $+11.4$] |
| QuaRot-W4A4 | 3 | 255 | $+1.2$ [$-6.6$, $+8.9$] [$-4.7$, $+6.3$] | $+6.3$ [$-8.4$, $+21.0$] [$-2.4$, $+14.1$] | $+9.4$ [$+1.7$, $+17.1$] [$+2.0$, $+16.5$] |
| SpinQuant-W16 path | 3 | 255 | $-0.4$ [$-2.1$, $+1.3$] [$-3.1$, $+2.4$] | $+2.0$ [$-1.4$, $+5.3$] [$-2.4$, $+6.3$] | $+2.4$ [$-0.6$, $+5.3$] [$-2.4$, $+7.1$] |
| SpinQuant-W4A4 | 3 | 255 | $+3.1$ [$-2.9$, $+9.2$] [$-2.0$, $+8.2$] | $+11.8$ [$+6.7$, $+16.8$] [$+5.1$, $+18.4$] | $+13.7$ [$+9.3$, $+18.2$] [$+5.5$, $+21.6$] |

Table 22: All q8 ordinary BF16-sourced states admitted by the panel rules, including unquantized path controls. Effects in pp; each cell: estimate with fixed-pool 95% $t$ interval, then crossed task×block 95% interval. All three modes use the same admitted states within a row. These descriptive intervals are unadjusted; the primary method family is P1–P3. KIVI-2 admits only two blocks after one-time infrastructure re-runs; it is not estimable under the original strict rule. Other cells admit three blocks. No action-only interval excludes zero; several takeover or joint-upgrade intervals do.

ery admitted configuration, with both fixed-pool and new-task intervals.

# G Exploratory Follow-ups: Label Transfer and Takeover Duration

Both analyses are exploratory, and no main-text claim rests on them. The label-transfer analysis is post hoc. The takeover analysis code was frozen before its outcomes were read, but its configurations were chosen after earlier results had been seen.

## G.1 Transfer of gain labels between upgrade modes

**Design.** Protocol `LABEL_TRANSFER_POSTHOC.md` (exploratory, post hoc: these outcomes had been seen). For a selection label $X$ and an executed mode $Y$, both in $\{G, H, F\}$, and a budget of 5, 10 or 20%, we select by $X$ and report the executed rescues, damages and net of $Y$, with ties broken uniformly at random (exact expectation). In the split-half variant, tasks are ranked by mean $X$ on one half of the blocks (outcome-independent hash) and evaluated on the other half, averaged over both directions; the diagonal is a self-label reference, not an upper bound, since it can also do poorly on the evaluation half. Same-draw selection on one set of outcomes is a hindsight diagnostic: its loss is non-negative by construction, so only its gain over random selection is informative. This tests known tasks across nonces, not new tasks or single decisions. Intervals are block and crossed task×block bootstrap, uncorrected for the 18

off-diagonal comparisons per frame. Budgets count states or tasks, not BF16 compute: $H$ and $F$ use several BF16 decisions per selected state.

**Results.** The protocol designates ordinary states (Bank U U1, 3 blocks) as the primary frame and first-divergence states (Bank P) as secondary, and designates no budget or label pair. In the primary frame no split-half loss is resolved ($H \to F$ at 10%: $+0.30$ [$-0.60$, $+1.11$]), and the self-label references themselves cover zero. In the secondary frame (W4 pool, split-half, pp of all units), at a 10% task budget, the losses with crossed intervals above zero are $H \to F$ 2.82 [1.40, 3.90], $H \to G$ 2.43 [0.97, 3.68] and $G \to F$ 1.50 [0.20, 2.19]; at 5%, $H \to G$ and $H \to F$ (1.70 each) also exclude zero, and at 20%, $H \to F$ 3.39 [1.94, 4.91]. The mode ranking implied by the selection label matches the self-label ranking for $F$ at 10 and 20% and for $G$ at 5% (point estimates). Same-draw results, U2B and KV8 pools are in the output file.

### G.2 Takeover duration

**Results.** Under SpinQuant-W4A4, against Q's own continuation, a one-decision takeover shows no detectable net gain (0.0, crossed [$-3.5$, $+3.5$]). Four decisions gain $+6.3$ pp: block-$t$ [$+0.2$, $+12.4$] (2 df), crossed [0.0, $+12.2$], so the crossed interval does not confirm it, and an all-state gain alone does not show that value is kept after handback to Q. Four decisions fall 5.5 pp short of sustained takeover ($+11.8$; shortfall block-$t$ [3.8, 7.2], crossed [0.8, 10.6]). Four-decision windows handed control back at 66% of states; the rest ended inside the window, so they cannot show that four steps suffice. Under GPTQ-W4 the continuation value itself is small ($+0.8$, crossed [$-5.9$, $+7.5$]) and no window effect is resolved. In the selected configuration takeover value depends on its length (cf. handoff costs, Ganz et al., 2026); GPTQ-W4 does not resolve whether this generalises.

**Design and reading.** Protocol `TAKEOVER_WINDOW_PROTOCOL.md` (tw-v1, exploratory; analysis code frozen before outcomes were read, Amendments 2–4). Both cells were chosen after earlier results were seen: GPTQ-W4 for its unresolved sustained-takeover value, SpinQuant-W4A4 for its large one ($H = +11.8$ pp on this panel, Table 22); $V_{Q,\infty}$ reuses that endpoint, so the SpinQuant results are conditional on this selection, and no interval is adjusted for this configuration selection. At 255 q8 BF16-sourced states (blocks f01–f03, 85 tasks each), the current action comes from Q (or B); BF16 executes the next $L$ complete decisions and hands back to Q. $V_{Q,L} = Y_{Q,L} - Y_{Q,0}$ is the value of an $L$-decision BF16 window after Q's action, $R_{Q,L} = Y_{Q,L} - Y_{Q,\infty}$ its shortfall against sustained takeover. All six tasks passed acceptance (0 errors, 0 overflows, 256/256 endpoint re-runs reproduced, nothing quarantined). The $E_{Q,4}$=1 subset keeps states where continuous BF16 generated more than four decisions, so a four-decision window really hands back. Intervals: block-$t$ over 3 block means (registered primary, df $= 2$) and crossed task×block bootstrap (Table 23). The frozen reading script uses the registered primary interval only; Amendment 3 requires at least two blocks and a nonzero-width interval. It reports the all-state SpinQuant four-decision gain as above zero and states that this alone is not value kept after handback, because states that end inside the window contribute $V_{Q,\infty}$ mechanically. The paper reports each takeover result under both intervals and does not call a result resolved when the crossed interval includes zero, a reading adopted when the results were read, not one fixed in the protocol. The four-decision gain is resolved under block-$t$ but not under the crossed interval both on all states and on the $E_{Q,4}$=1 subset. Differences between action sources at each $L$ ($\Delta_L$) are unresolved in both cells.

## H Supplementary Analyses and Claims Not Supported by Their Controls

No main-text claim rests on this appendix. It reports a supplementary injection panel and claims that we withdrew because their controls did not support them.

**Controlled injection.** On 30 ordinary GPTQ-W4 states per model, each from one development block, we compared a forced invalid action with a matched valid action and with a process perturbation that repeated the last successful read; either B or Q then continued. Each panel balanced wrong-ID, tool-call/schema error and premature-write perturbations. Table 24 gives the invalid-minus-process contrast. These small panels did not resolve an error-specific effect; the q8 interval under Q still allows substantial harm.

Replay checks passed on both panels; q4 lacked three recorded audit fields. In q4, most wrong-tool injections actually produced schema errors

| cell | states | $V_{Q,1}$ | $V_{Q,4}$ | $V_{Q,\infty}$ | $R_{Q,4}$ |
|---|---|---|---|---|---|
| GPTQ-W4 | all 255 | +2.4<br>$t$: [-5.4, +10.1]<br>crossed: [-3.1, +7.8] | +2.0<br>$t$: [-5.4, +9.3]<br>crossed: [-3.9, +8.2] | +0.8<br>$t$: [-5.3, +6.9]<br>crossed: [-5.9, +7.5] | +1.2<br>$t$: [-6.6, +8.9]<br>crossed: [-3.1, +6.3] |
| GPTQ-W4 | $E_{Q,4}$=1 | +3.4<br>$t$: [-8.9, +15.7]<br>crossed: [-4.7, +10.9] | +2.0<br>$t$: [-7.0, +11.0]<br>crossed: [-6.3, +10.6] | +0.2<br>$t$: [-8.2, +8.6]<br>crossed: [-9.9, +9.5] | +1.8<br>$t$: [-10.6, +14.1]<br>crossed: [-5.1, +9.6] |
| SpinQuant-W4A4 | all 255 | 0.0<br>$t$: [-2.9, +2.9]<br>crossed: [-3.5, +3.5] | +6.3<br>$t$: [+0.2, +12.4]<br>crossed: [0.0, +12.2] | +11.8<br>$t$: [+6.7, +16.8]<br>crossed: [+5.1, +18.4] | −5.5<br>$t$: [-7.2, -3.8]<br>crossed: [-10.6, -0.8] |
| SpinQuant-W4A4 | $E_{Q,4}$=1 | −1.8<br>$t$: [-6.2, +2.7]<br>crossed: [-5.5, +1.6] | +4.2<br>$t$: [+1.2, +7.2]<br>crossed: [-3.0, +11.6] | +12.5<br>$t$: [+7.2, +17.7]<br>crossed: [+4.6, +21.0] | −8.3<br>$t$: [-10.9, -5.6]<br>crossed: [-16.0, -1.2] |

Table 23: Takeover duration (pp; [block-$t$; crossed] 95%). SpinQuant-W4A4 four-decision windows handed back at 66% of states (169/255) and generated about 228 BF16 tokens per state, against about 60 for one-decision windows.

| Model | continuation | error minus process [95%] | failed checks |
|---|---|---|---|
| q4 | B | 0.00 [−0.13, +0.13] | 2/30 |
| q4 | Q | −0.03 [−0.20, +0.10] | 2/30 |
| q8 | B | −0.03 [−0.13, +0.07] | 9/30 |
| q8 | Q | −0.13 [−0.30, +0.03] | 9/30 |

Table 24: Error-specific contrast, in success probability units; task-cluster bootstrap. Failed manipulation checks are state counts, repeated here for the two continuations of the same panel. No equivalence claim follows from intervals including zero.

(8/10). In q8, seven premature writes failed to inject the intended write (six environment rejections and one execution without a database change), and two valid references returned "not found". These failures prevent the panels from establishing a general recovery mechanism.

### H.1 Claims not supported by their controls

**W×KV interaction (not supported).** A first q4 analysis found a W4×KV4 interaction of $-14.9$ $[-23.8, -6.0]$ pp ($n_{\text{blk}} = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is $-9.0$ $[-30.6, +12.5]$. The pre-registered q8 replication gives $+2.1$ $[-5.8, +10.1]$ over 6 blocks, crossed $[-8.2, +12.4]$. The replication is not supported, and we make no W×KV claim.

**Static precision restoration.** Restoring attention or a layer segment was compared with byte-matched random modules. In the damaged RTN-W3 models the attention contrast was $+9.8$ $[+6.1, +13.6]$ pp (crossed $[+4.5, +15.1]$), while the last third was $-8.6$ $[-13.4, -3.8]$ (crossed $[-12.9, -4.7]$). These were exploratory contrasts among 39 unadjusted comparisons; the attention reference had only two completed random draws. None of the 30 GPTQ-W3 or W4 contrasts excluded zero under both intervals, and no tested allocation beat whole-model GPTQ-W4 at matched storage. This incomplete stress test supports no deployment method. The full matrix and unfinished-arm inventory remain in the released analysis.

**Provenance.** Figure values, retained tables and scoring definitions map to committed analysis outputs and frozen manifests. The release also retains complete pattern counts, restoration matrices, process labels, experiment logs and historical corrections; they are not additional independent evidence for the paper's conclusions. Protocol amendments and unresolved primary contrasts remain disclosed in this supplement. Removing duplicate displays and exploratory leads from this supplement does not change the estimators, admission rules or reported primary decisions.